Frequency–Spatial Dual-Path Modeling for Remote Sensing Image Semantic Segmentation

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***Abstract*—Semantic segmentation of remote sensing images (RSIs) faces persistent challenges in high-frequency information degradation, boundary ambiguity, and structural inconsistency. To address these issues, we present a frequency–spatial dual-path modeling network (FSDPNet) paradigm that achieves holistic structural enhancement and fine-grained detail preservation. Specifically, we introduce a frequency-reduction Mamba mechanism (FRMM), which explicitly disentangles and adaptively retains critical high-frequency and directional representations during encoding. In the decoding stage, a multiscale frequency-driven reconstruction module (MFRM) leverages frequency-domain enhancement and dilated convolutions for progressive resolution recovery and precise texture compensation. Furthermore, a segment anything model (SAM)-constrained semantic consistency fusion module (SCFM) harnesses structural priors to reinforce boundary delineation and suppress semantic ambiguity across scales. The overall training is guided by a composite objective that jointly optimizes global semantic consistency and local structural fidelity. Extensive experiments on multiple remote sensing benchmarks demonstrate that our approach consistently outperforms state-of-the-art segmentation methods, setting a new standard for structural integrity and fine-detail preservation in RSI segmentation. All codes and segmentation results will be available at: https://github.com/darkseid-arch/FSDPNet**

***Index Terms*—Boundary delineation, remote sensing, semantic consistency, semantic segmentation, structure feature.**

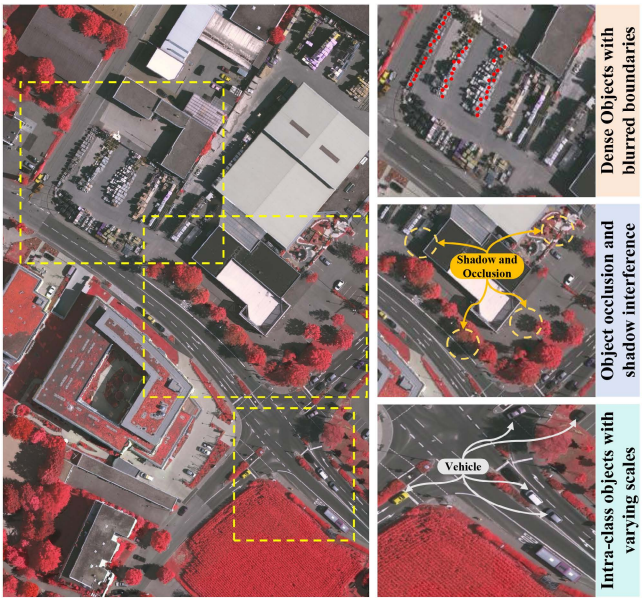


Fig. 1. Illustration of challenges in RSI semantic segmentation. (Left) Original scene, with yellow boxes highlighting typical problem areas. (Right) Zoomed-in view to show: (top) dense objects with blurred boundaries, (middle) object occlusion and shadow interference, and (bottom) intraclass objects with varying scales.

I. Introduction

SEMANTIC segmentation of remote sensing images (RSIs) is a core technique for geospatial information extraction, supporting applications such as land use survey [1], environmental monitoring [2], urban planning [3], disaster assessment [4], and military reconnaissance [5]. Advances in high-resolution sensors have greatly enhanced the spatial detail and structural complexity of RSIs, enabling fine-grained classification and automatic interpretation. However, as illustrated in Fig. 1, RSIs present unique challenges compared with natural images, including multiscale and multisource data, strong background interference, and complex surface reflectance [6]. Subtle spectral and textural differences between land cover types, as well as frequent boundary ambiguity caused by dense object distributions, occlusions, shadows, and transmission artifacts [7], significantly hinder accurate segmentation. Furthermore, the prevalence of small objects, elongated structures, and layered spatial arrangements [8] increases the difficulty in achieving scale-adaptive segmentation and consistent boundary delineation. Consequently, reconciling boundary and semantic detail with global structural consistency [9] remains a central challenge in RSI semantic segmentation.

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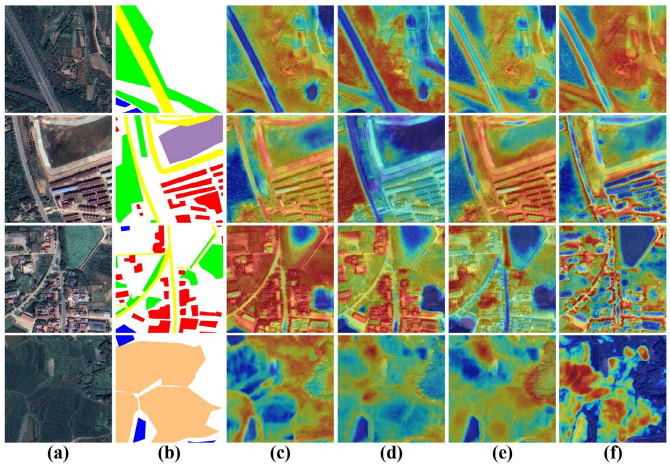


Fig. 2. Feature visualizations of different semantic segmentation methods on RSIs. (a) Original image. (b) Ground truth. (c) CNN-based. (d) Transformer-based. (e) Mamba-based. (f) Our method.

Early approaches to RSI semantic segmentation mainly relied on hand-crafted features and traditional image processing methods for pixel- or region-level classification [10], [11], [12]. While these techniques achieved moderate results in simple scenes, they struggled with generalizability, multiscale adaptation, and complex spatial structures [13], [14]. With the rise of deep learning, convolutional neural networks (CNNs) have become the mainstream framework for their automatic feature learning and multiscale representation [15], [16]. However, the locality of CNNs limits their ability to capture long-range dependencies, leading to challenges in boundary precision and global semantic consistency. Recently, Transformer-based models have advanced the field by leveraging global self-attention to model cross-scale relationships, yet they often suffer from high computational cost and inadequate local detail modeling [17], [18]. To overcome these limitations, a new class of sequence modeling architectures, such as Mamba and its variants, has emerged, providing efficient global modeling with lower computational overhead [19], [20]. These advances mark a promising step toward more accurate and robust semantic segmentation in complex scenarios.

Although CNN-, Transformer-, and Mamba-based models have driven significant progress in RSI segmentation, existing approaches still struggle to balance global semantic consistency and fine-grained structural detail, particularly in complex scenes, as shown in Fig. 2. On one hand, the progressive downsampling and limited receptive fields of CNNs, as well as the global attention mechanisms of Transformers and sequence modeling of Mamba, tend to compromise high-frequency cues and subtle boundary information. This typically results in blurred or over-smoothed object edges, missing fine structures, and difficulties in precisely delineating densely distributed or small-scale targets. On the other hand, methods focusing on global context often fail to preserve local structural fidelity, leading to errors in both large-scale object classification and the segmentation of intricate boundaries. Furthermore, most current models lack dedicated mechanisms for adaptively extracting and maintaining fine-grained features across varying object scales, making it challenging to achieve accurate segmentation in the presence of complex spatial arrangements and ambiguous boundaries. As a result, the field still faces critical challenges in encoding and preserving high-frequency information, extracting complete fine-grained details, mitigating boundary smoothing, and maintaining a coherent balance between global and local representations.

To address the problem of RSI semantic segmentation, this article presents a frequency–spatial dual-path modeling network (FSDPNet) framework, integrating multiscale and multilevel feature optimization. The core design introduces a frequency-reduction Mamba mechanism (FRMM) in the main encoding path, which leverages a segment anything model (SAM)-derived [21] semantic priors and coordinate attention to adaptively preserve high-frequency information and alleviate detail loss caused by repeated downsampling. To further enhance spatial detail and boundary delineation, a multiscale frequency-driven reconstruction module (MFRM) is incorporated, enabling progressive restoration of fine-grained features across multiple scales via multidepth frequency encoding and cross-scale feature interaction. In addition, an SAM-constrained semantic consistency fusion module (SCFM) aggregates multilevel structure cues under semantic guidance, reinforcing boundary clarity and promoting global–local feature optimization. Through the coordinated integration of these modules, the proposed FSDPNet framework achieves robust performance in both global semantic consistency and fine-grained segmentation. The main contributions are as follows.

1) We propose an FSDPNet framework that jointly exploits frequency and spatial domain representations, enabling comprehensive multiscale and multilevel feature optimization for remote sensing semantic segmentation.
2) We design an FRMM that leverages semantic priors and coordinate attention to adaptively preserve high-frequency details and mitigate local loss during encoding, thereby enhancing the retention of spatial details.
3) We introduce an MFRM to restore fine-grained spatial features and refine boundaries via multidepth frequency encoding and cross-scale feature interaction, leading to improved segmentation accuracy in complex scenes.
4) We develop an SCFM that aggregates multilevel structural cues under semantic guidance, thereby reinforcing boundary clarity and promoting robust global–local semantic consistency and segmentation performance.

Section II reviews the related work. Section III presents the proposed FSDPNet framework and its associated components. Section IV shows the experimental analysis, and Section V provides the discussion. Section VI concludes this work.

II. Related Works

A. Semantic Segmentation of RSIs

The proliferation of high-resolution remote sensing imagery has continuously raised the bar for semantic segmentation models, demanding improvements in structural awareness, scale adaptivity, and boundary delineation. CNNs have traditionally dominated this field due to their robust capability for local feature extraction. Early efforts, such as DDCM-Net [22], RDU-Net [23], and ResUNet-a [24], explored multibranch convolutional structures, hierarchical feature aggregation, and

hybrid designs to strengthen contextual understanding and object localization. To overcome challenges related to scale variation and boundary ambiguity, various position- and scale-aware attention mechanisms, as well as boundary refinement strategies, have been introduced [25], [26], [27]. These developments have led to notable gains in the delineation of fine structures and small objects. Nevertheless, the inherent locality of standard CNNs limits their ability to capture global semantic context and long-range dependencies, which are crucial for accurate segmentation in complex scenes. Recent advances in Transformer-based architectures have substantially alleviated these limitations, enabling comprehensive global context modeling and multiscale feature fusion. Hybrid CNN-Transformer frameworks, in particular, have demonstrated significant performance improvements, especially in scenarios involving intricate spatial distributions and indistinct boundaries [28], [29], [30], [31]. However, these sophisticated models often come with high computational costs and may still struggle with precise boundary recovery. To address these challenges, state-space models (SSMs), and especially the Mamba architecture, have emerged as efficient alternatives for long-range dependency modeling and global context aggregation [19], [32], [33], [34]. Integrating Mamba modules with CNNs or Transformers further enhances multiscale representation and boundary sensitivity, while maintaining computational efficiency. Representative approaches such as MBSSNet [35] and HMAFNet [36] leverage cross-domain fusion to deliver robust multiscale segmentation, and collaborative modeling strategies are increasingly adopted in remote sensing applications.

B. Sam for RSI Semantic Segmentation

The emergence of foundation models has fundamentally reshaped visual segmentation, providing a unified framework for a wide range of tasks. SAM marks a breakthrough in universal segmentation by enabling prompt-driven extraction of arbitrary object masks [21]. Leveraging a Vision Transformer (ViT)-based image encoder [37] for global feature representation, a prompt encoder for integrating user guidance, and a mask decoder for generating segmentation masks, SAM demonstrates strong generalization across diverse visual domains. Building on this foundation, numerous variants have been tailored for remote sensing applications. Architectures such as FastSAM [38], MobileSAM [39], SAM-HQ [40], and Semantic-SAM [41] introduce lightweight designs, architectural enhancements, or semantic constraints to address the unique challenges of remote sensing imagery, including large-scale variance and complex object boundaries. The release of large-scale datasets like SA-1B has further facilitated the adaptation of SAM-based models to remote sensing tasks. Despite these advances, SAM's performance on RSIs is often limited by domain gaps, significant scale variations, and intricate scene composition [42], [43], [44]. To address these issues, methods such as RingMoSAM [42] utilize multimodal data for improved mask quality, while RSPrompter [43] employs task-aware prompt learning to enhance model adaptability. Data-centric approaches, exemplified by SSRS [44], focus on domain-specific training and optimized prompt annotation strategies. Generative techniques, including SGO and SGB-based frameworks, further bolster generalization by leveraging synthetic data augmentation [45], [46]. Nonetheless, challenges remain regarding prompt robustness, accurate delineation of structures, and computational efficiency [47].

C. Image Processing in Frequency Domain

Frequency-domain learning has become an influential paradigm in image analysis, providing models with the capacity to leverage spectral cues for enhanced feature representation and discrimination [48], [49]. Techniques based on Fourier and discrete cosine transforms (DCTs) are particularly effective at capturing global structural patterns and addressing the inherent limitations of spatial-domain approaches [50], [51]. Recent advancements highlight the applicability of frequency-domain techniques beyond remote sensing, such as in medical image segmentation [52], [53], underwater semantic segmentation [54], and real-time autonomous driving [55]. Within semantic segmentation, frequency self-attention mechanisms have been introduced to simultaneously enhance edge fidelity and reduce computational overhead, consistently delivering state-of-the-art results across benchmark datasets [56]. DCT-based decomposition facilitates the explicit separation of high- and low-frequency components, thereby supporting multiscale semantic interpretation in remote sensing imagery. Cross-domain attention modules, such as frequency–channel attention and frequency-guided fusion, enable adaptive aggregation of spectral and channelwise signals, substantially improving model robustness under challenging imaging conditions [57]. Joint spatial–spectral self-attention frameworks further promote comprehensive feature fusion and effective global context modeling, while hierarchical aggregation of spatial and frequency features yields more precise boundary delineation [58], [59]. Spectral–spatial fusion strategies and adaptive frequency modules have shown particular efficacy in enhancing segmentation reliability amid class imbalance and complex backgrounds [60]. Frequency-guided edge enhancement, coupled with frequency-weighted objectives, further sharpens boundary localization and segmentation consistency [61]. Moreover, multifrequency fusion networks integrate information from diverse spectral subbands, achieving superior performance across a range of segmentation tasks [62]. These developments underscore the value of frequency-domain modeling in semantic segmentation, establishing a robust and versatile framework for integrating spectral and spatial information in image understanding.

III. Methodology

To address the challenges of multiscale feature representation, frequency information loss during downsampling, and boundary ambiguity in remote sensing semantic segmentation, we propose the FSDPNet framework. As illustrated in Fig. 3, FSDPNet adopts a dual-path encoder–decoder architecture, incorporating an FRMM, an SAM-constrained SCFM, and an MFRM. These components collaboratively mitigate frequency–spatial information loss and semantic inconsistency, while enhancing feature discrimination and detail preservation. In our pipeline, the SAM-derived semantic priors are generated offline for each training image in a preprocessing stage via the SAM.

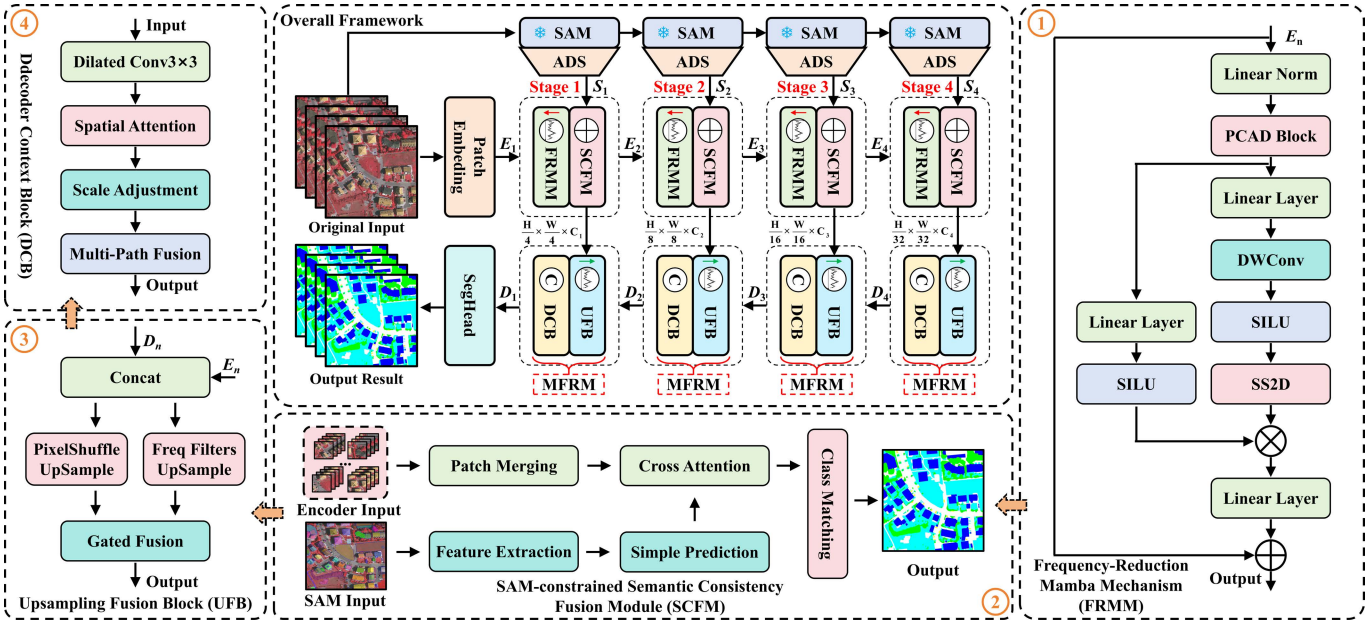


Fig. 3. Overall structure of the proposed FSDPNet. The architecture employs a dual-path encoder–decoder framework, comprising an FRMM, an SAM-constrained SCFM, and an MFRM. Semantic priors from SAM guide the encoding process, while multilevel feature fusion and frequency–spatial interactions collaboratively enhance segmentation accuracy and detail preservation in remote sensing imagery.

Algorithm 1 FSDPNet Inference Pipeline

Input: Input image $I \in \mathbb{R}^{H\times W\times 3}$
Output: Segmentation probability map $\hat{Y} \in [0,1]^{H\times W\times C}$

Obtain semantic prior masks M_{SAM} using $\text{SAM}(I)$;
for $s = 1$ ***to*** 4 **do** // Encoder stages
  Apply FRMM: $F_s \leftarrow \text{FRMM}(F_{s-1})$;
  Obtain SAM mask via adaptive downsampling: $\bar{M}_s \leftarrow \text{ADS}(M_{\text{SAM}}, s)$;
  Fuse via SCFM: $F_s^{\text{fused}} \leftarrow \text{SCFM}(F_s, \bar{M}_s)$;
for $s = 4$ ***down to*** 1 **do** // Decoder stages
  Fuse skip connection F_s^{fused} and upsampled feature;
  $D_s \leftarrow \text{MFRM}(\cdot)$;
  Integrate via UFB, DCB;
Aggregate $\{D_1, \ldots, D_4\}$;
Predict segmentation: $\hat{Y} \leftarrow \text{SegHead}(\text{Concat}(D_1, \ldots, D_4))$;
return $\hat{Y}$

These masks are solely used as auxiliary supervision during training to enhance semantic learning in the encoder and are not required at inference or deployment time. Thus, during inference, only the raw input image $I \in \mathbb{R}^{H\times W\times 3}$ is provided to FSDPNet, with no dependence on SAM or precomputed priors. Each stage of the main encoder path applies FRMM for frequency-aware downsampling and attention-based feature refinement, suppressing redundant low-frequency content while maintaining semantic clarity. The auxiliary path adapts the offline-generated SAM masks via adaptive downsampling (ADS), aligning them with main-path features at each scale. Features from both paths are fused through SCFM, where semantic consistency constraints reinforce class discrimination. Four encoder stages produce multiscale features, which are processed in the decoder via skip connections and multifrequency fusion. Within the decoder, MFRM performs frequency enhancement and spatial detail reconstruction, while the upsampling fusion block (UFB) and decoder context block (DCB) integrate multiscale context and spatial attention, effectively reducing boundary blurring and object confusion. Finally, the aggregated hierarchical features $\{F_1, \ldots, F_4\}$ are passed to the SegHead, yielding the segmentation probability map $\hat{Y} \in [0,1]^{H\times W\times C}$. Algorithm 1 illustrates the detailed workflow of the proposed FSDPNet framework.

A. Frequency-Reduction Mamba Mechanism

In high-resolution remote sensing semantic segmentation, repeated downsampling and complex scene composition often amplify high-frequency noise, particularly along object boundaries and in regions with fine structures [22], [23], [24]. This can lead to semantic confusion, detail loss, and blurred boundaries. To address these issues, we introduce the FRMM, which enables controllable frequency feature modulation during encoding. FRMM effectively suppresses aliasing and noise, while maximizing the retention of discriminative information. As shown in Fig. 4, given an input feature map $X \in \mathbb{R}^{H\times W\times C}$, we first project X into the frequency domain using a 2-D DCT as follows:

$$\mathbf{F}(u,v) = \alpha(u)\,\alpha(v)\,\mathbf{X}(x,y) \times \cos\left(\frac{\pi(2x+1)u}{2H}\right)\cos\left(\frac{\pi(2y+1)v}{2W}\right) \quad (1)$$

where $\alpha(\cdot)$ is a normalization factor. This spectral representation enables the separation of frequency components, facilitating selective frequency modulation. To suppress noise

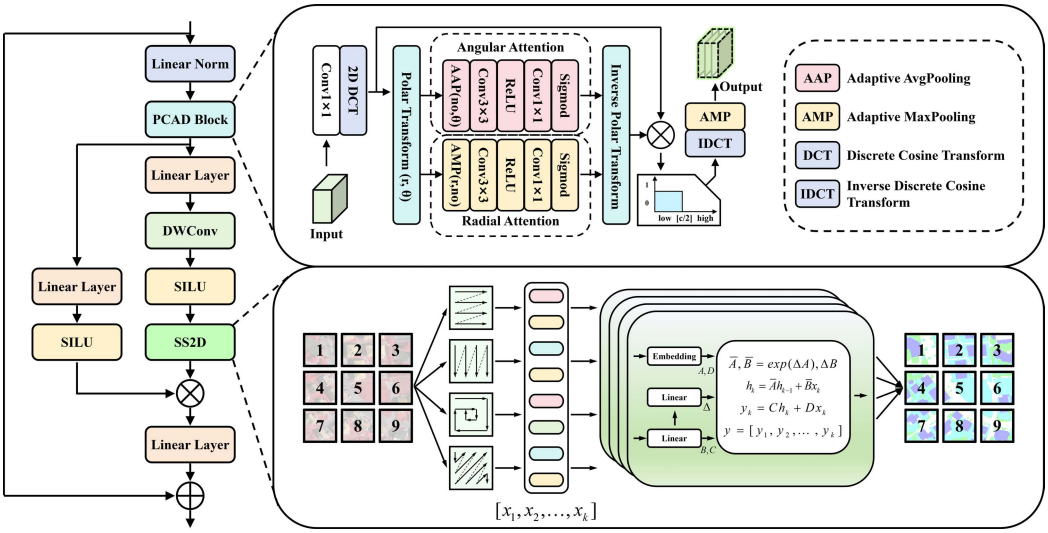


Fig. 4. Architecture of the proposed FRMM. The module projects features into the frequency domain via 2-D DCT, applies adaptive frequency masking, and performs polar transformation with angular and radial attention to selectively enhance salient components.

and retain salient information, we apply an adaptive frequency mask $\mathbf{M}(u, v)$ to filter high-frequency and noisy components

$$\hat{\mathbf{F}}(u, v) = \mathbf{M}(u, v) \odot \mathbf{F}(u, v). \tag{2}$$

To further capture directional and spatial context, we transform the frequency map into polar coordinates

$$\mathbf{F}_r(\theta, r) = \hat{\mathbf{F}}(r\cos\theta, r\sin\theta). \tag{3}$$

The angular and radial attention mechanisms are then introduced to adaptively aggregate directional and distance-aware frequency features as follows:

$$\mathbf{A}_\theta = \sigma\left(\mathbf{W}_\theta \cdot \delta\left(\mathrm{Conv}_{3\times3}\left(\mathrm{AAP}\left(\mathbf{P}\right)\right)\right)\right) \tag{4}$$

$$\mathbf{A}_r = \sigma\left(\mathbf{W}_r \cdot \delta\left(\mathrm{Conv}_{3\times3}\left(\mathrm{AMP}\left(\mathbf{P}\right)\right)\right)\right) \tag{5}$$

where AAP$(\cdot)$ and AMP$(\cdot)$ denote the adaptive average and max pooling, respectively. The aggregated feature is fused as

$$\mathbf{P}'(r, \theta) = \mathbf{P}_r(r, \theta) \odot \mathbf{A}_\theta(\theta) \odot \mathbf{A}_r(r). \tag{6}$$

The resulting polar-domain features are then mapped back to Cartesian coordinates and reconstructed into the spatial domain via inverse DCT

$$Y(x, y) = \sum_{u=0}^{H-1}\sum_{v=0}^{W-1} \alpha(u, v)\ \tilde{\mathbf{F}}(u, v) \times \cos\left(\frac{\pi(2x+1)u}{2H}\right)\cos\left(\frac{\pi(2y+1)v}{2W}\right). \tag{7}$$

This process enables fine-grained suppression of irrelevant frequencies and enhancement of salient structures, preserving spatial details while reducing aliasing and information loss during encoding. To further enhance global context modeling and computational efficiency, we embed the filtered frequency features into a lightweight Mamba state-space module

$$h_k = Ah_{k-1} + Bx_k, \quad y_k = Ch_k + Dx_k \tag{8}$$

where x_k and h_k denote the input and hidden states, respectively. The Mamba block excels at capturing long-range dependencies and high-order interactions, ensuring that frequency-refined features retain discriminative power across encoder stages and contribute to robust multiscale representation.

B. SAM-Constrained SCFM

For semantic segmentation, although deep encoders can efficiently capture global and contextual representations, they often struggle to preserve semantic consistency and boundary accuracy, especially around complex object edges. To address these challenges, we propose the SAM-constrained SCFM, which leverages structure-aware priors from the SAM [21] to guide the integration of boundary and semantic cues. As shown in Fig. 5, given an input encoder feature map $\mathbf{F}_{\mathrm{enc}} \in \mathbb{R}^{H\times W\times C}$ and the corresponding SAM-derived boundary prior $\mathbf{F}_{\mathrm{sam}} \in \mathbb{R}^{H\times W\times C}$, we first extract prominent boundary information and suppress noise using a ReLU-activated convolution as follows:

$$U_{\mathrm{enc}} = \mathrm{ReLU}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}_{\mathrm{enc}}\right)\right). \tag{9}$$

For the SAM branch, a GeLU-activated convolution preserves weak object cues and stabilizes gradient propagation

$$V_{\mathrm{sam}} = \mathrm{GeLU}\left(\mathrm{Conv}_{3\times3}\left(\mathbf{F}_{\mathrm{sam}}\right)\right). \tag{10}$$

To enhance feature stability and facilitate consistent fusion, layer normalization is applied to V_{sam}

$$\hat{V}_{\mathrm{sam}} = \mathrm{LayerNorm}\left(V_{\mathrm{sam}}\right). \tag{11}$$

Next, we introduce a channel attention block (CAB) to adaptively recalibrate feature responses and emphasize semantically salient regions. The CAB output is formulated as

$$F_{\mathrm{sam}} = \hat{V}_{\mathrm{sam}} \odot \sigma\left(W_2\ \mathrm{ReLU}\left(W_1\ \mathrm{GAP}\left(\hat{V}_{\mathrm{sam}}\right)\right)\right) \tag{12}$$

where GAP$(\cdot)$ denotes the global average pooling, W_1 and W_2 are the learnable projection matrices, and σ is the sigmoid activation. This formulation enables the module to enhance boundary features while suppressing redundant background responses. For semantic-guided spatial alignment, we employ

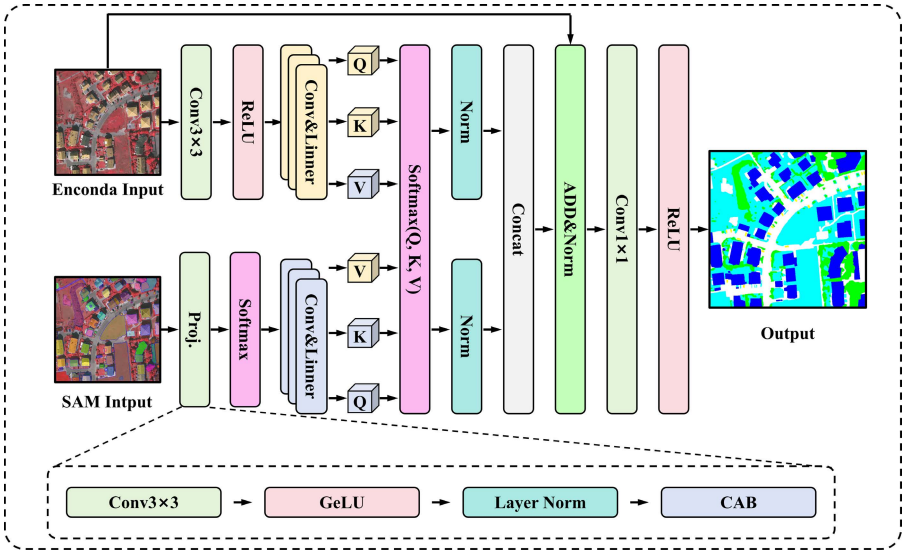


Fig. 5. Architecture of the proposed SAM-constrained SCFM. The module integrates encoder features and SAM-derived boundary priors via nonlinear transformation, channel attention, and cross-attention mechanisms. Layer normalization and feature fusion ensure both spatial semantic consistency and boundary precision, enabling robust and fine-grained segmentation in complex remote sensing scenes.

a cross-attention mechanism. Encoder and SAM features are projected to query, key, and value spaces

$$Q_{\text{enc}} = U_{\text{enc}} W_q^e, \quad K_{\text{sam}} = F_{\text{sam}} W_k^s, \quad V_{\text{sam}} = F_{\text{sam}} W_v^s \tag{13}$$

$$A_{e\to s} = \text{Softmax}\left(\frac{Q_{\text{enc}} K_{\text{sam}}^T}{\sqrt{d}}\right) \tag{14}$$

where $A_{e\to s}$ encodes the attention between encoder features and SAM priors, adaptively weighting the contribution of boundary information at each spatial location. Similarly, we compute attention from SAM features to encoder features as $A_{s\to e}$. The enhanced features are then updated as

$$F'_{\text{enc}} = A_{e\to s} V_{\text{sam}} \tag{15}$$

$$F'_{\text{sam}} = A_{s\to e} V_{\text{enc}}. \tag{16}$$

We concatenate and normalize the updated features and compress them via a 1×1 convolution with ReLU activation

$$F_{\text{fuse}} = \text{ReLU}\left(\text{Conv}_{1\times 1}\left(\text{Norm}\left(F'_{\text{enc}} \| F'_{\text{sam}}\right)\right)\right) \tag{17}$$

where $\|$ denotes the channelwise concatenation and Norm$(\cdot)$ is a normalization operation to ensure feature stability. The fused output F_{fuse} serves as an enhanced representation, preserving both semantic consistency and fine boundary localization.

C. Multiscale Frequency-Driven Reconstruction Module

The key challenge in the decoding stage lies in simultaneously ensuring the progressive recovery of segmentation accuracy and the fine restoration of high-frequency details. Conventional upsampling methods often suffer from boundary blurring and detail loss, especially in complex scenes such as building rooftops and intricate road edges.

To address these issues, as shown in Fig. 6, we propose the MFRM, which integrates an UFB and a DCB. Specifically, at each decoding stage, the encoding features $\mathbf{E}_n \in \mathbb{R}^{H_n \times W_n \times C}$ are concatenated with the output features $\mathbf{D}_n$ from the previous decoding layer

$$\mathbf{F}_n = \text{Concat}\left(\mathbf{E}_n, \mathbf{D}_n\right). \tag{18}$$

A 1×1 convolution with ReLU activation is first applied for channel compression as follows:

$$\mathbf{F}'_n = \sigma\left(\mathbf{W}_{1\times 1}\mathbf{F}_n + \mathbf{b}_1\right) \tag{19}$$

where $\sigma(\cdot)$ denotes the ReLU function. In the frequency enhancement branch, $\mathbf{F}'_n$ is transformed into the frequency domain using a 2-D discrete Fourier transform (DFT)

$$\hat{\mathbf{F}}_n(u,v) = \sum_{x=0}^{H_n-1}\sum_{y=0}^{W_n-1} \mathbf{F}'_n(x,y)\, e^{-j2\pi\left(\frac{ux}{H_n}+\frac{vy}{W_n}\right)}. \tag{20}$$

The frequency spectrum is decomposed into multiband components, enabling selective enhancement of high-frequency structures and suppression of noise. For different frequency bands, adaptive modulation is performed

$$\hat{\mathbf{F}}_n^{(k)}(u,v) = \mathbf{M}^{(k)}(u,v) \odot \hat{\mathbf{F}}_n(u,v), \quad k = 1,\ldots,K \tag{21}$$

where $\mathbf{M}^{(k)}$ are the learnable bandpass filters. Each band is then refined via a frequency-aware transformation as $\tilde{\mathbf{F}}_n^{(k)} = \mathbf{W}_f^{(k)} \hat{\mathbf{F}}_n^{(k)}$. The fused frequency features are projected back to the spatial domain using the inverse DFT

$$\mathbf{F}''_n = \frac{1}{H_n W_n}\sum_{u=0}^{H_n-1}\sum_{v=0}^{W_n-1} \tilde{\mathbf{F}}_n(u,v)\, e^{j2\pi\left(\frac{ux}{H_n}+\frac{vy}{W_n}\right)}. \tag{22}$$

This frequency-to-spatial domain conversion enables precise recovery of both regular and irregular details while preventing high-frequency artifacts often introduced by naive upsampling. To further refine spatial alignment and edge localization, PixelShuffle is employed

$$\mathbf{U}_n = \text{PixelShuffle}\left(\mathbf{F}''_n\right). \tag{23}$$

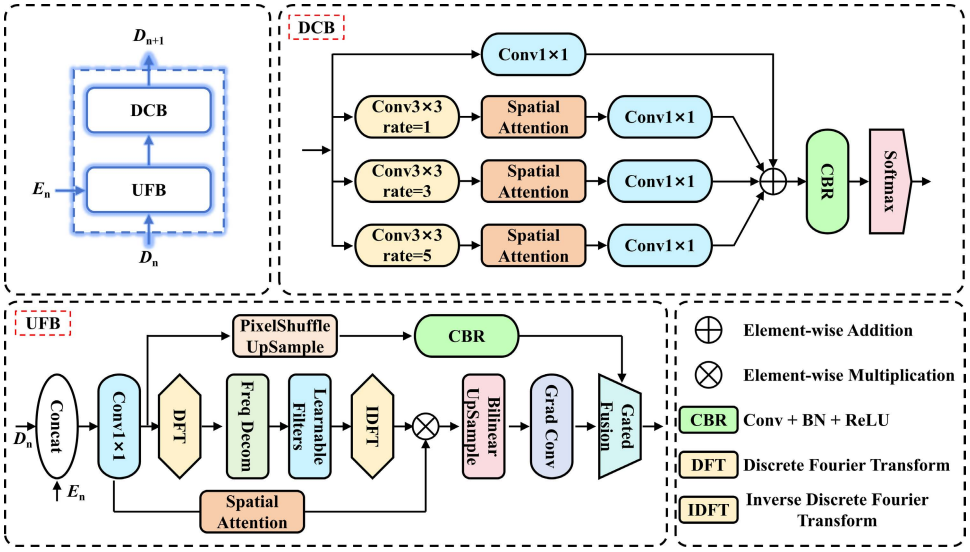


Fig. 6. Framework of the proposed MFRM, which integrates frequency-domain enhancement (UFB) and multiscale spatial context aggregation (DCB) to progressively restore fine boundaries and semantic details during decoding.

A global average pooling (GAP) branch is introduced to aggregate multiscale semantic cues

$$\mathbf{U}_n' = \mathbf{U}_n \odot \sigma\left(\mathbf{W}_g \text{AvgPool}\left(\mathbf{U}_n\right) + \mathbf{b}_g\right). \tag{24}$$

Finally, the DCB captures extensive multiscale contextual information using atrous convolutions with varying dilation rates and spatial attention

$$\mathbf{R}_n^{(r)} = \sigma\left(\mathbf{W}_r\left(\mathbf{U}_n' + \mathbf{U}_n^{(r)}\right) + \mathbf{b}_r\right), \quad r \in \{1,3,5\} \tag{25}$$

where $\mathbf{U}_n^{(r)}$ denotes the output of a 3×3 convolution with dilation rate r. The fused output is aggregated using a CBR block (Conv-BN-ReLU)

$$\mathbf{D}_{n+1} = \text{CBR}\left(\sum_r \mathbf{R}_n^{(r)}\right). \tag{26}$$

This hierarchical decoding process not only enhances segmentation accuracy at each stage but also ensures the effective recovery of high-frequency boundaries and multiscale context.

D. Multicomponent Joint Optimization Loss

To simultaneously improve overall classification accuracy, detail preservation, and semantic consistency of boundaries, we design a multicomponent joint optimization loss function

$$\mathcal{L}_{\text{total}} = \lambda_{CE}\mathcal{L}_{CE} + \lambda_F\mathcal{L}_{\text{freq}} + \lambda_{SLM}\mathcal{L}_{SLM} \tag{27}$$

where λ_{CE}, λ_F, and λ_{SLM} are the weights for each loss term, with $\lambda_{CE} + \lambda_F + \lambda_{SLM} = 1$.

1) Pixelwise Cross-Entropy Loss: The standard pixelwise cross-entropy loss L_{CE} encourages correct per-pixel classification and is defined as

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log p_{i,c} \tag{28}$$

where N is the number of pixels, C is the number of classes, $y_{i,c} \in \{0,1\}$ is the one-hot ground-truth label for pixel i and class c, and $p_{i,c}$ is the predicted probability.

1) Frequency Consistency Loss: To ensure the restoration of high-frequency details and spatial semantic consistency, we introduce a frequency consistency loss L_{freq} based on the 2-D DFT

$$\mathcal{L}_{\text{freq}} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathcal{F}\left(P_i\right) - \mathcal{F}\left(G_i\right)\right\|_2^2 \tag{29}$$

where $\mathcal{F}(\cdot)$ denotes the 2-D DFT, P_i and G_i are the prediction and ground-truth patches centered at pixel i. This loss constrains the frequency components to enhance the recovery of fine structural details and suppress high-frequency artifacts, improving edge and boundary delineation.

2) Semantic Line-Sensitive Loss: To further enhance boundary precision and semantic consistency, we introduce a semantic line-sensitive loss L_{SLM} that adaptively weights the loss for pixels near semantic boundaries. The pixelwise boundary weight m_j is calculated as follows:

$$m_j = \sigma\left(\alpha\left|\nabla_x G_j\right| + \beta\left|\nabla_y G_j\right|\right) \tag{30}$$

where ∇_x and ∇_y denote the horizontal and vertical gradient operators on the ground truth G_j, α and β are the directional weighting coefficients, and $\sigma(\cdot)$ is the sigmoid function. This design ensures higher sensitivity to boundary regions during optimization. The line-sensitive loss is then formulated as

$$\mathcal{L}_{SLM} = -\frac{1}{M}\sum_{j=1}^{M}\sum_{c=1}^{C} m_j\, y_{j,c}\log p_{j,c} \tag{31}$$

where M is the number of pixels considered, and m_j emphasizes pixels in boundary regions.

IV. Experiments and Analysis

A. Dataset Description

To comprehensively evaluate the effectiveness of the proposed FSDPNet for semantic segmentation in remote sensing imagery, experiments were conducted on the ISPRS datasets (Potsdam and Vaihingen) [73] and the LoveDA dataset [74]. The details of each dataset are as follows.

1) ISPRS Potsdam Dataset: This benchmark provides a large-scale, high-resolution aerial imagery collection widely used for remote sensing tasks. The dataset covers six land cover classes: impervious surfaces, buildings, low vegetation, trees, cars, and clutter. The original images have a spatial resolution of 5 cm and a size of 6000 × 6000 pixels. For model training and evaluation, 14 images are used for training and 24 for testing. The imagery is provided in three channels: Red, Green, and Blue. To fit the input size of deep models, the images are randomly cropped into nonoverlapping patches of 512 × 512 pixels, resulting in 140 training tiles.

2) ISPRS Vaihingen Dataset: The Vaihingen dataset is collected under similar protocols as Potsdam, with high-resolution aerial images covering the same six semantic classes. Compared with Potsdam, Vaihingen features more complex urban layouts with higher building density and more intricate boundaries, posing greater challenges for semantic segmentation. The original images have a spatial resolution of 9 cm and a size of 2500 × 2500 pixels. The dataset consists of 33 images, with 17 for training and 16 for testing. All images are divided into 24 nonoverlapping patches of 512×512 pixels.

3) LoveDA Dataset: This dataset is a large-scale, high-resolution semantic segmentation dataset focused on urban-rural domain adaptation. The images are acquired from three representative urban and rural areas in China, with significant scene diversity and pronounced domain gaps, making it highly valuable for real-world applications. The dataset provides RGB images with a spatial resolution of 0.3 m and a patch size of 1024 × 1024 pixels. It includes seven annotated semantic classes: building, road, water, bare land, farmland, forest, and others. The entire dataset contains 2522 training images, 1669 validation images, and 1796 test images.

B. Implementation Details

All experiments were conducted on a high-performance computing platform equipped with an Intel[1] Xeon[1] CPU E5-2683 v4 and an NVIDIA RTX A5000 GPU (24-GB VRAM), running Ubuntu 20.04. The software environment included Python 3.10, PyTorch 1.13.1, and CUDA 12.3. During model training, to address the limited number of annotated samples in remote sensing datasets, we employed various data augmentation strategies, including random cropping, horizontal/vertical flipping, and color jittering, to improve model generalization. Furthermore, we adopted a transfer learning approach by initializing the backbone network with weights pretrained on large-scale datasets and fine-tuning them for the target task. For model initialization, we used the He initialization method [75] to ensure stable convergence. The Adam optimizer was employed with an initial learning rate of 1×10^{-4} and a weight decay of 5×10^{-3} to mitigate overfitting. To enhance learning rate adaptability, a cosine annealing schedule [76] was applied throughout the 300 training epochs. The input image size was uniformly set to 512 × 512 pixels for both training and testing to maintain consistency. The mini-batch size was set to 8, and training was performed on a single GPU. For inference, all models received input images at the same size as in training, and pixelwise predictions were generated. For evaluation, we report results using ground truth with per-pixel annotations, and all quantitative metrics are computed at the original image resolution to ensure fair and comprehensive comparisons.

[1]Registered trademark.

C. Evaluation Metrics

We evaluate model performance using three standard metrics: overall accuracy (OA), mean $F1$-score (m$F1$), and mean intersection over union (mIoU). The OA reflects the proportion of correctly classified pixels

$$\mathrm{OA} = \frac{TP + TN}{TP + FP + FN + TN}. \tag{32}$$

The m$F1$ metric averages the $F1$-score across all classes, balancing precision and recall

$$\mathrm{F1}_i = \frac{2 \cdot \mathrm{Precision}_i \cdot \mathrm{Recall}_i}{\mathrm{Precision}_i + \mathrm{Recall}_i} \tag{33}$$

$$\mathrm{mF1} = \frac{1}{C} \sum_{i=1}^{C} \mathrm{F1}_i \tag{34}$$

$$\mathrm{Precision}_i = \frac{TP_i}{TP_i + FP_i}, \quad \mathrm{Recall}_i = \frac{TP_i}{TP_i + FN_i}. \tag{35}$$

The mIoU measures the average overlap between predicted and ground-truth regions for all classes

$$\mathrm{IoU}_i = \frac{|P_i \cap G_i|}{|P_i \cup G_i|}, \quad \mathrm{mIoU} = \frac{1}{C} \sum_{i=1}^{C} \mathrm{IoU}_i \tag{36}$$

where TP, TN, FP, and FN are the true/false positive/negative counts, P_i and G_i are the predicted and ground-truth sets for class i, and C is the total number of classes.

D. Comparison With State-of-the-Art Methods

To demonstrate the superiority of the proposed FSDPNet, we conduct comprehensive performance comparisons on three benchmark datasets. The compared methods include CNN-based (FarSeg [63], ABCNet [64], HCANet [65], MANet [69], and LOGCAN++ [72]), Transformer-based (Segmenter [66], DC-Swin [67], UNetFormer [68], and RSSFormer [70]), and Mamba-based (RS3Mamba [71] and UNetMamba [19]) methods. The detailed experimental results and analysis are as follows.

1) Experimental on ISPRS Potsdam: Table I presents a quantitative comparison between FSDPNet and state-of-the-art semantic segmentation methods on the ISPRS Potsdam test set. As shown, our method achieves superior performance across all key metrics, including OA, mIoU, and m$F1$. Specifically, FSDPNet attains an OA of 95.16%, mIoU of 79.04%, and m$F1$ of 88.30%, outperforming the best CNN-based baseline (LOGCAN++) by 3.9%, 0.46%, and 1.68%, respectively, and surpassing the latest Mamba-based method (UNetMamba) by 4.8%, 2.39%, and 0.66%. For per-class IoU, FSDPNet consistently achieves the highest or near-highest scores across all categories, including "impervious surface" (88.29%), "building" (93.91%), "tree" (81.22%), "car" (93.76%), and "clutter" (91.06%), demonstrating exceptional accuracy and robustness, particularly in challenging

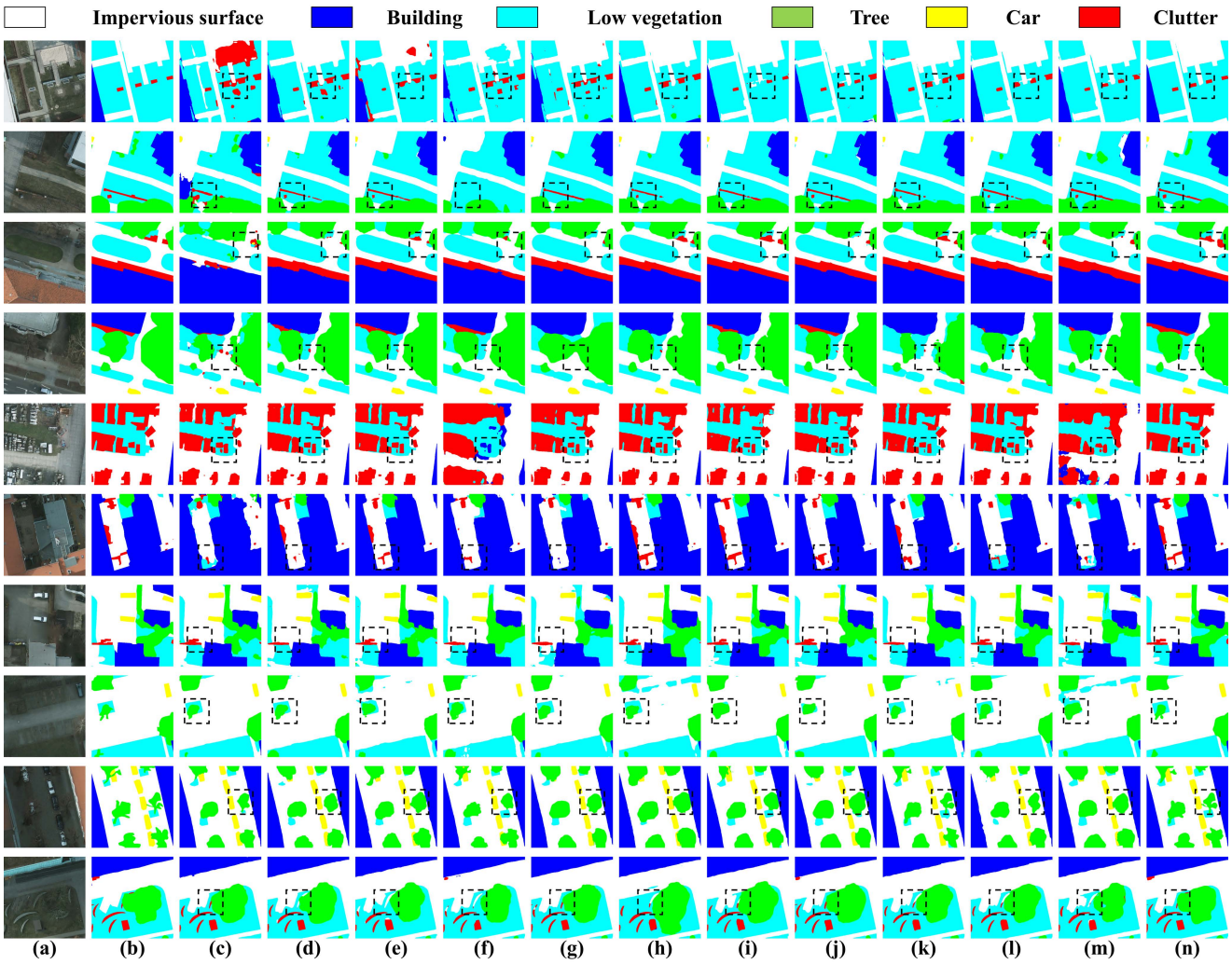


Fig. 7. Qualitative comparison of segmentation results on the Potsdam test set. (a) Input image. (b) Ground truth. (c) FarSeg. (d) ABCNet. (e) HCANet. (f) Segmenter. (g) DC-Swin. (h) UNetFormer. (i) MANet. (j) RSSFormer. (k) RS3Mamba. (l) LOGCAN++. (m) UNetMamba. (n) FSDPNet (Ours).

TABLE I

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE ISPRS POTSDAM TEST SET. BEST RESULTS IN EACH COLUMN ARE BOLDED

Methods	Backbone	Imp-suf	Building	Low-veg	Tree	Car	Clutter	OA (%)	mIoU (%)	mF1 (%)
FarSeg [65]	ResNet-50	86.93	93.56	76.72	79.00	91.37	40.32	84.90	77.98	86.29
ABCNet [66]	ResNet-50	84.78	92.02	76.52	78.49	91.70	36.39	89.92	76.65	85.21
HCANet [67]	ResNet-50	86.41	81.55	71.44	70.77	74.00	35.64	89.44	69.97	85.43
Segmenter [68]	Swin-B	80.54	86.12	70.69	68.35	77.34	26.60	77.91	68.27	79.17
DC-Swin [69]	Swin-B	84.88	90.47	74.95	76.42	89.16	34.83	82.80	75.12	84.18
UNetFormer [70]	ResNet-50	84.81	91.40	74.34	77.54	91.77	35.51	89.38	76.06	84.85
MANet [71]	ResNet-50	84.49	91.48	76.36	78.12	91.03	**41.17**	89.81	77.11	85.85
RSSFormer [72]	RSS-B	86.92	93.12	77.19	79.55	91.43	39.75	89.91	78.05	86.01
RS3Mamba [73]	R18-VMamba-T	84.61	92.85	73.42	76.40	91.51	39.75	89.84	77.85	86.04
LOGCAN++ [74]	ResNet-50	87.51	93.76	77.20	79.78	93.13	40.10	91.06	78.58	86.62
UNetMamba [75]	ResT-Lite	85.75	93.16	75.68	79.02	92.02	39.62	90.36	76.65	87.64
FSDPNet (Ours)	FRMamba	**88.29**	**93.91**	**78.35**	**81.22**	**93.76**	38.68	**95.16**	**79.04**	**88.30**

regions such as building boundaries and small-object classes like cars. Qualitative results are illustrated in Fig. 7, where FSDPNet produces more continuous and precise segmentation boundaries compared with other methods. CNN-based approaches (e.g., FarSeg, ABCNet, HCANet, and MANet) often suffer from boundary blurring and fragmentation in complex urban scenes. Transformer-based models (e.g., Segmenter, DC-Swin, UNetFormer, and RSSFormer) improve global consistency but tend to over-smooth fine details, especially for small and thin structures. Mamba-based methods (e.g., RS3Mamba and UNetMamba) reduce boundary artifacts to some extent, but still struggle with complex backgrounds.

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE ISPRS VAIHINGEN TEST SET. BEST RESULTS IN EACH COLUMN ARE BOLDED

Methods	Backbone	Imp-suf	Building	Low-veg	Tree	Car	Clutter	OA (%)	mIoU (%)	mF1 (%)
FarSeg [65]	ResNet-50	85.96	91.55	71.95	80.63	78.24	33.69	80.45	73.67	83.20
ABCNet [66]	ResNet-50	84.86	89.72	69.14	79.85	76.80	45.81	89.49	74.37	84.44
HCANet [67]	ResNet-50	86.68	79.95	70.04	69.38	72.55	**49.38**	87.83	71.33	82.83
Segmenter [68]	Swin-B	78.69	82.17	65.03	75.85	34.18	15.03	67.12	58.49	70.07
DC-Swin [69]	Swin-B	83.39	87.76	69.34	78.89	59.07	18.42	73.69	66.15	76.65
UNetFormer [70]	ResNet-50	85.46	91.16	70.26	79.46	74.27	35.44	80.00	72.67	82.70
MANet [71]	ResNet-50	85.37	90.50	70.18	80.23	80.23	41.49	89.90	74.65	84.37
RSSFormer [72]	RSS-B	86.28	91.25	72.49	79.36	74.00	41.49	81.00	73.36	83.12
RS3Mamba [73]	R18-VMamba-T	86.62	91.83	67.84	83.66	81.97	36.14	81.72	75.03	83.72
LOGCAN++ [74]	ResNet-50	85.87	91.85	70.35	83.39	80.14	38.63	81.94	74.72	82.94
UNetMamba [75]	ResT-Lite	87.20	94.76	68.82	82.21	**82.21**	35.21	91.91	77.11	84.16
FSDPNet (Ours)	FRMamba	**88.27**	**95.05**	**74.28**	**83.92**	81.98	43.04	**92.57**	**77.76**	**85.94**

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE LOVEDA TEST SET. BEST RESULTS IN EACH COLUMN ARE BOLDED

Method	Backbone	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU (%)
FarSeg [65]	ResNet-50	45.18	56.55	56.92	79.78	14.96	46.59	62.50	51.78
ABCNet [66]	ResNet-50	52.02	43.84	33.92	60.91	17.32	36.62	52.44	42.60
HCANet [67]	ResNet-50	46.07	57.15	54.93	76.84	42.16	46.31	60.43	54.84
Segmenter [68]	Swin-B	39.70	44.71	47.29	73.51	8.35	37.45	55.71	43.82
DC-Swin [69]	Swin-B	42.34	50.17	54.83	77.13	15.86	42.76	59.28	47.81
UNetFormer [70]	ResNet-50	44.16	56.51	55.84	79.00	13.86	44.27	62.65	50.90
MANet [71]	ResNet-50	38.56	59.06	52.58	67.88	35.63	37.38	36.02	46.73
RSSFormer [72]	RSS-B	44.03	57.23	55.35	79.37	10.91	45.22	62.04	50.59
RS3Mamba [73]	R18-VMamba-T	41.60	58.23	54.03	77.34	17.97	43.81	61.37	50.62
LOGCAN++ [74]	ResNet-50	47.37	58.38	56.46	80.05	18.44	47.91	64.80	51.35
UNetMamba [75]	ResT-Lite	55.32	59.43	55.41	79.12	49.03	49.75	63.75	58.83
FSDPNet (Ours)	FRMamba	**58.27**	**62.77**	**58.35**	**80.60**	37.24	**50.26**	**66.08**	**59.08**

In contrast, FSDPNet effectively preserves the integrity and sharpness of architectural edges, tree crowns, and vehicle contours. This improvement is mainly attributed to the FRMM module, which adaptively enhances high-frequency boundary information, and the SCFM and MFRM modules, which jointly optimize semantic consistency and multiscale feature representation. As a result, FSDPNet achieves the best overall balance between global accuracy and local detail preservation on the ISPRS Potsdam dataset.

2) Experimental on ISPRS Vaihingen: Table II reports the quantitative performance comparison between FSDPNet and a variety of mainstream semantic segmentation methods on the ISPRS Vaihingen test set. As shown, FSDPNet achieves the highest or second-highest IoU in nearly all classes, especially for challenging categories such as impervious surface (88.27%), building (95.05%), low vegetation (74.28%), tree (83.92%), and car (81.98%). Compared with the best-performing baselines, FSDPNet improves the mIoU and m$F1$ by approximately 2.0% and 1.8%, respectively, and also achieves the highest OA (OA = 92.57%).

These results demonstrate the superior ability of FSDPNet to preserve high-frequency details and fine boundaries across multiple scales. Fig. 8 presents qualitative segmentation results on the Vaihingen test set. Traditional CNN-based methods (e.g., FarSeg, ABCNet, HCANet, MANet, and LOGCAN++) still exhibit obvious boundary confusion and fragmentation in complex regions, such as building corners, road intersections, and tree–low vegetation boundaries. Transformer-based methods (e.g., Segmenter, DC-Swin, UNetFormer, and RSSFormer) provide some advantages in global context modeling, but tend to over-smooth high-frequency details, leading to the loss of small structures such as cars and thin boundaries. Mamba-based methods (e.g., RS3Mamba and UNetMamba) perform better in modeling long-range dependencies, but still struggle to preserve fine details of small targets. In contrast, FSDPNet achieves an optimal balance between global consistency and local detail preservation, producing more accurate and sharper results, especially for cars, road edges, and building boundaries. The superior segmentation performance of FSDPNet in these challenging scenarios further highlights its effectiveness, particularly in preserving roof edges of buildings and small cars and suppressing boundary confusion in complex scenes.

3) Experimental on LoveDA: Table III presents the performance comparison between FSDPNet and various state-of-the-art segmentation methods on the LoveDA test set. As shown, FSDPNet achieves significant improvements in several key categories, outperforming all competing methods in background (58.27%), building (62.77%), road (58.35%), water (80.60%), and agriculture (66.08%) IoU scores. Notably, FSDPNet attains the highest mIoU of 59.08%, surpassing the previous best method UNetMamba (58.83%), further validating its strong generalization ability in complex and

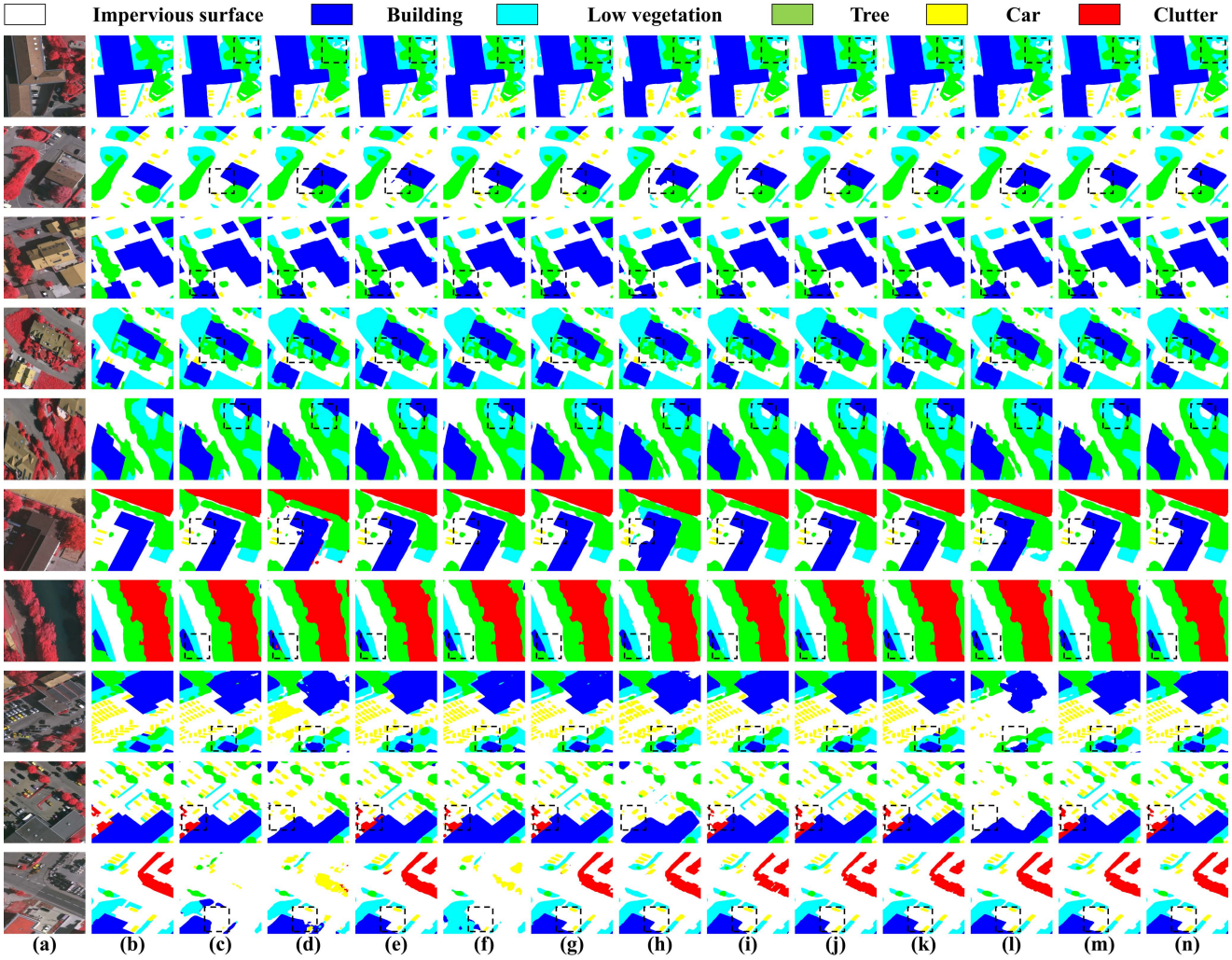


Fig. 8. Qualitative comparison of segmentation results on the Vaihingen test set. (a) Input image. (b) Ground truth. (c) FarSeg. (d) ABCNet. (e) HCANet. (f) Segmenter. (g) DC-Swin. (h) UNetFormer. (i) MANet. (j) RSSFormer. (k) RS3Mamba. (l) LOGCAN++. (m) UNetMamba. (n) FSDPNet (Ours).

heterogeneous scenes. Among the compared methods, CNN-based approaches (e.g., FarSeg, ABCNet, HCANet, MANet, and LOGCAN++) tend to perform inconsistently, especially for rare classes such as Barren and in regions with complex boundaries. Transformer-based methods (e.g., Segmenter, DC-Swin, UNetFormer, and RSSFormer) show better global modeling capability but often fail to capture fine-grained structures along roads, forests, and similar boundaries. In contrast, FSDPNet leverages the complementary effects of FRMM and MFRM modules to alleviate feature mixing across multiple scales, effectively decoding high-frequency details and preserving spatial consistency. This leads to more accurate and refined segmentation, particularly in challenging categories and complex edge regions. Fig. 9 provides qualitative comparisons, where FSDPNet produces more complete and coherent results, with improved preservation of object boundaries and fine structural details. Specifically, in the visual results, CNN-based methods often suffer from severe fragmentation and boundary ambiguity in roads, agricultural, and building regions. Transformer-based approaches generally achieve better continuity, but still struggle with scattered buildings, small-scale forests, and thin targets. Mamba-based methods (e.g., RS3Mamba and UNetMamba) improve long-range modeling but may lose spatial detail in highly variable urban–rural transitions. In comparison, FSDPNet consistently maintains global structural coherence while significantly enhancing the completeness and sharpness of local boundaries, demonstrating its clear advantages in the LoveDA benchmark.

4) Experimental on RS-RGB-T: We further evaluate the effectiveness of frequency-domain modeling on the RS-RGB-T dataset [82], which consists of 4024 urban road aerial images with dense pixel-level labels for nine semantic categories.

Table IV reports the comparative results of FSDPNet and several representative frequency-based segmentation baselines. Our FSDPNet achieves the best OA (89.76%), mean IoU (72.48%), and mean $F1$-score (83.94%), outperforming SFFNet [77], FDNet [78], AFENet [79], FGSwin [80], and FreDNet [81] across nearly all categories. In particular, FSDPNet shows significant improvements in challenging classes such as motorcycle (59.55% IoU), human (47.38%), and truck (87.67%), clearly exceeding the runner-up models. These gains demonstrate the effectiveness of our frequency–spatial dual-path approach in capturing both coarse scene structure and fine-grained object details. Fig. 10 presents visual

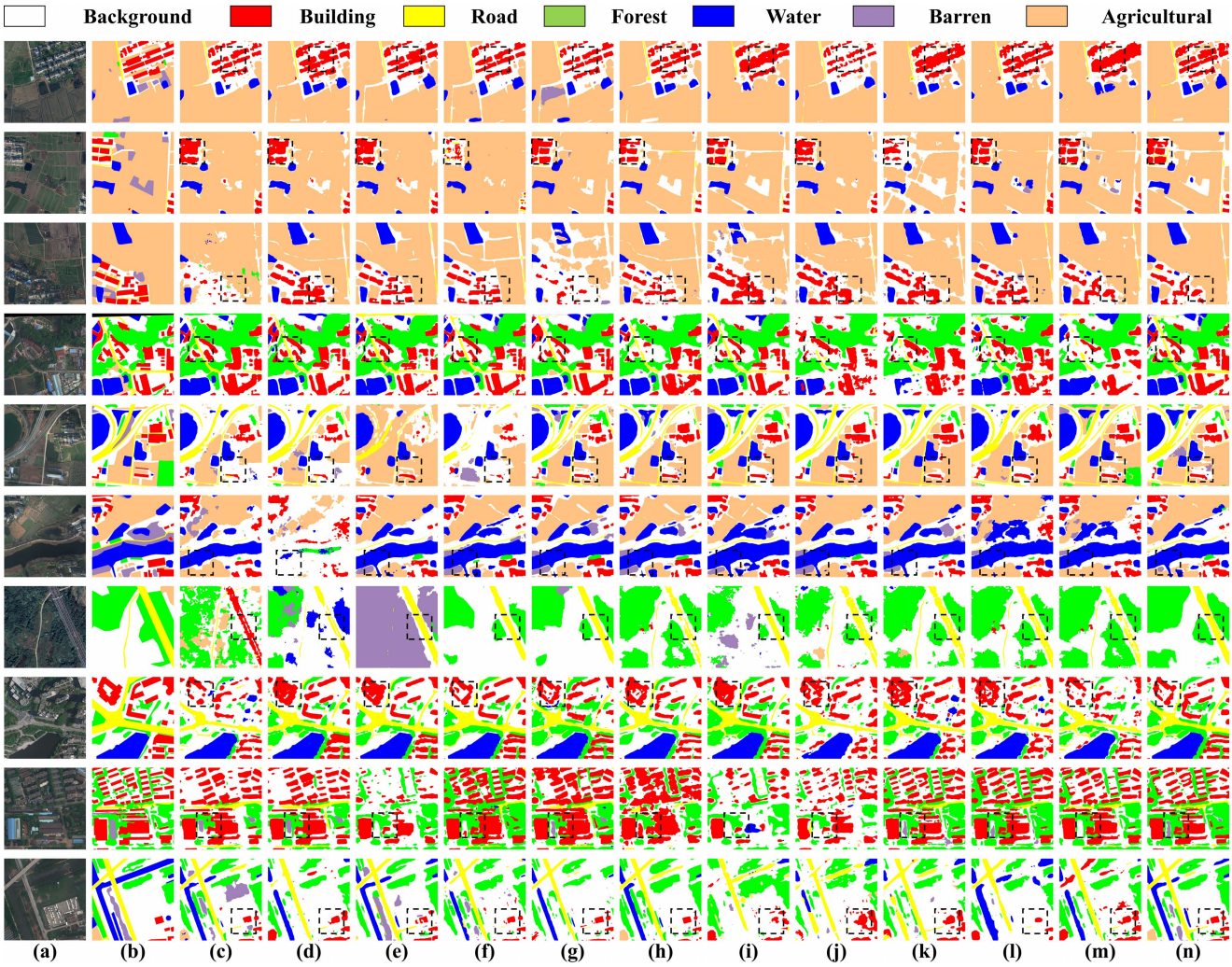


Fig. 9. Qualitative comparison of segmentation results on the LoveDA test set. (a) Input image. (b) Ground truth. (c) FarSeg. (d) ABCNet. (e) HCANet. (f) Segmenter. (g) DC-Swin. (h) UNetFormer. (i) MANet. (j) RSSFormer. (k) RS3Mamba. (l) LOGCAN++. (m) UNetMamba. (n) FSDPNet (Ours).

TABLE IV

COMPARISON WITH STATE-OF-THE-ART FREQUENCY-BASED METHODS ON THE RS-RGB-T TEST SET. BEST RESULTS IN EACH COLUMN ARE BOLDED

Methods	Road	Building	Motorcycle	Car	Tree	Human	Traffic	Truck	OA (%)	mIoU (%)	mF1 (%)
SFFNet [80]	84.97	85.15	53.71	76.83	71.24	37.11	44.80	78.41	80.92	66.03	76.79
FDNet [81]	81.34	81.02	42.18	70.92	68.27	29.98	34.09	68.84	77.32	59.08	72.84
AFENet [82]	86.44	86.09	54.22	79.56	76.44	40.22	51.97	80.73	84.35	69.21	78.80
FGSwin [83]	78.49	76.11	36.71	60.12	70.77	35.15	28.34	74.26	72.81	57.11	66.43
FreDNet [84]	84.01	83.90	46.23	72.02	67.81	34.20	39.17	67.26	77.98	61.82	74.29
FSDPNet (Ours)	**87.95**	**88.62**	**59.55**	**82.34**	**78.93**	**47.38**	**55.42**	**87.67**	**89.76**	**72.48**	**83.94**

comparisons under various lighting and scene conditions. FSDPNet generates more accurate, detailed, and continuous segmentation maps for both typical (e.g., daytime roads) and difficult cases (e.g., nighttime and occlusion). In particular, our method excels at delineating small objects, such as motorcycles and pedestrians, and handling clutter or low-illumination areas, where other frequency-based baselines often produce incomplete or noisy results. FSDPNet shows superior boundary localization and preserves fine structures, effectively reducing false positives and fragmentations. The RS-RGB-T experiments demonstrate the robust generalization capability of FSDPNet, validating the advantage of its dual-path frequency–spatial modeling in multimodal, high-complexity urban environments. Compared with other frequency-domain segmentation architectures, FSDPNet is particularly effective in scenarios with dynamic lighting, heavy occlusion, and small target density, highlighting its potential for real-world remote sensing applications.

5) Verification on Large-Scale Images: To further assess the practical generalization and robustness of the proposed framework in real-world remote sensing scenarios, we conduct qualitative comparisons between FSDPNet and other state-of-the-art methods on the original large-scale images

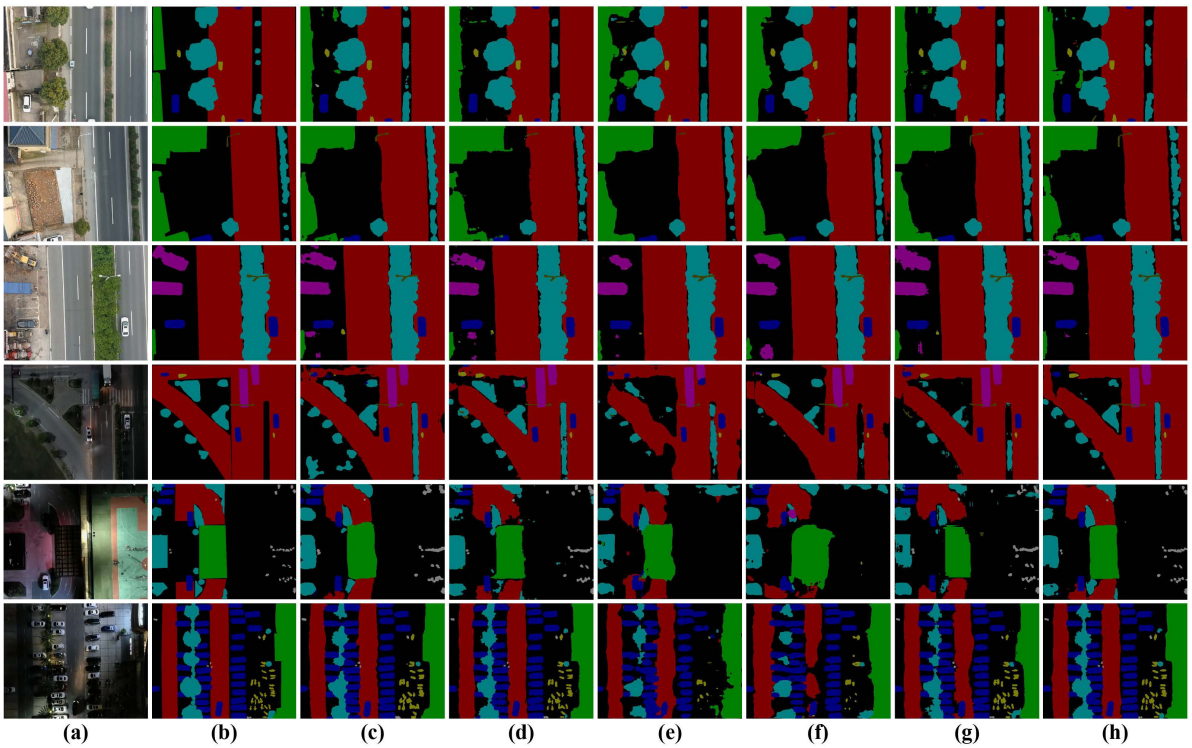


Fig. 10. Qualitative comparison of segmentation results on the RS-RGB-T test set. (a) Input image. (b) Ground truth. (c) SFFNet. (d) FDNet. (e) AFENet. (f) FGSwin. (g) FreDNet. (h) FSDPNet (Ours).

(2500 × 2500 pixels) from the ISPRS Potsdam test set. Large-scale images exhibit higher spatial resolution and more complex spatial distributions of ground objects, with pronounced long-range dependencies and richer multiscale object and boundary structures. These characteristics pose stricter demands on models' abilities for global semantic consistency, cross-scale feature construction, and preservation of high-frequency details and boundaries. As shown in Fig. 11, CNN-based models (e.g., FarSeg, ABCNet, HCANet, MANet, and LOGCAN++) tend to suffer from obvious boundary fragmentation, texture confusion, and suboptimal recognition of small targets in large-area images. Transformer-based models (e.g., Segmenter, DC-Swin, UNetFormer, and RSSFormer) generally maintain better semantic consistency over large regions, but still exhibit limitations in preserving high-frequency details and the fidelity of local boundaries. Mamba-based models (e.g., RS3Mamba and UNetMamba) further enhance long-range dependency modeling; however, they may still lose spatial granularity and precision in complex object transitions. In contrast, FSDPNet fully leverages the advantages of dual-path design, effectively suppressing feature mixing and aliasing across scales via frequency- and spatial-domain modules. This allows the model to accurately reconstruct intricate boundary and texture details, as well as to capture multiscale semantic information.

6) Analysis of Computational Complexity: Fig. 12 provides a comprehensive comparison of FSDPNet and other state-of-the-art methods on the ISPRS Potsdam test set in terms of FLOPs, parameter count, mIoU, and inference speed (frames/s). FLOPs and parameters vary widely among different architectures, reflecting diverse design philosophies and efficiency-accuracy tradeoffs. Among all methods, ABCNet and UNetFormer exhibit the lowest computational cost, with only 11.68G and 11.74G FLOPs, and 15.63M and 11.66M parameters, respectively, making them favorable for resource-constrained settings but with clear performance tradeoffs in complex scenarios. In contrast, Transformer-based models such as HCANet and DC-Swin have significantly higher resource demands, each exceeding 124G FLOPs and 118M parameters, due to their global modeling mechanisms. Notably, all reported inference metrics for FSDPNet (including frames/s, GPU memory consumption, and total runtime do not include any SAM prior generation, as SAM masks are only used during training and not at inference. Therefore, the efficiency results reflect the true end-to-end deployment performance of FSDPNet. As shown in Fig. 12, there is a clear tradeoff between mIoU, model complexity, and runtime efficiency. Our FSDPNet achieves a balanced outcome, offering competitive accuracy (79.04% mIoU) and high inference speed (28.6 frames/s), highlighting its potential for practical real-world applications.

E. Ablation Studies

To further validate the effectiveness of our model design, we conduct comprehensive ablation studies on three benchmarks. The analysis includes: 1) evaluating the contribution of each module; 2) effectiveness analysis of different module combinations; 3) performing sensitivity analysis regarding loss function weighting; and 4) impact of frequency transform choice. All experiments are conducted with identical training protocols to ensure fairness and reproducibility.

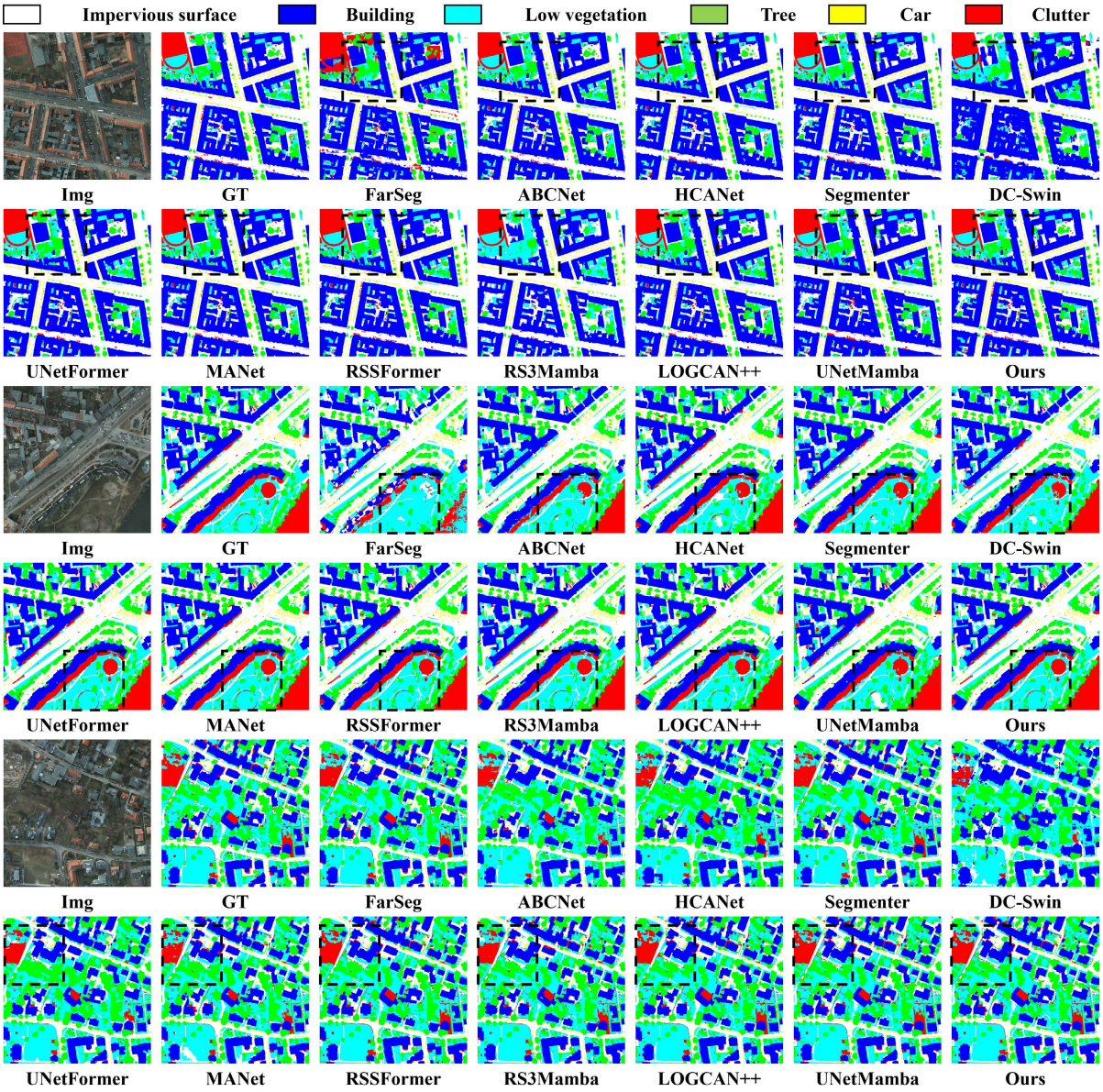


Fig. 11. Qualitative comparison of FSDPNet with state-of-the-art methods on large-scale images (2500 × 2500 pixels) from the ISPRS Potsdam test set.

1) Effectiveness of FRMM: As shown in Tables V–VII, introducing only the FRMM (without MFRM or SCFM) into the baseline architecture yields notable performance gains across all three datasets. Specifically, on the ISPRS Potsdam dataset, FRMM alone achieves an mIoU of 75.98%, representing an absolute improvement of 2.3% over the baseline. On the ISPRS Vaihingen and LoveDA datasets, the mIoU reaches 71.86% and 55.10%, respectively. These results demonstrate that FRMM is particularly effective in alleviating high-frequency information loss during downsampling. The impact of different frequency transforms is further evaluated in Tables VIII. Qualitative comparisons (see Figs. 13) further show that FRMM helps preserve structural consistency for roads, rooftops, and other high-frequency regions. Feature visualization (see Fig. 14) illustrates that FRMM enhances the semantic discrimination of high-frequency details, improving the model's ability to capture boundaries and fine-grained spatial structures.

2) Effectiveness of SCFM: As presented in Tables V–VII, using SCFM alone achieves mIoU scores of 76.50%, 70.88%, and 53.91% on the three datasets, outperforming the FRMM-only models. This indicates that SCFM plays a crucial role in enhancing edge sensitivity and semantic consistency. Notably, on the ISPRS Vaihingen dataset, SCFM improves the segmentation of challenging classes such as trees and low-vegetation by 2%–3% IoU, highlighting the effectiveness of SAM-based edge enhancement. Qualitative and feature visualizations (see Figs. 13–15) show that SCFM provides more uniform and prominent edge responses, reducing "leakage" and "blurring" around roads and small objects. This module demonstrates advantages for high-resolution RSIs with complex backgrounds and ambiguous boundaries.

3) Effectiveness of MFRM: MFRM is designed to enhance decoding by reconstructing multiscale spatial details. As

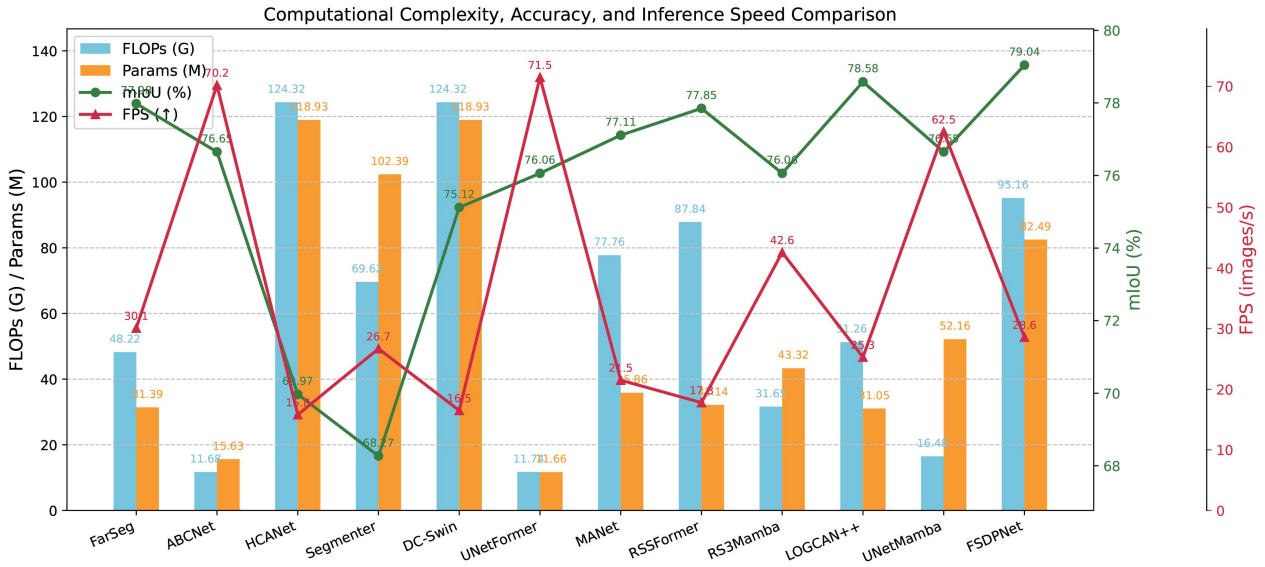


Fig. 12. Comparison of FLOPs, parameter counts, and mIoU (%) of different segmentation methods on the ISPRS Potsdam dataset.

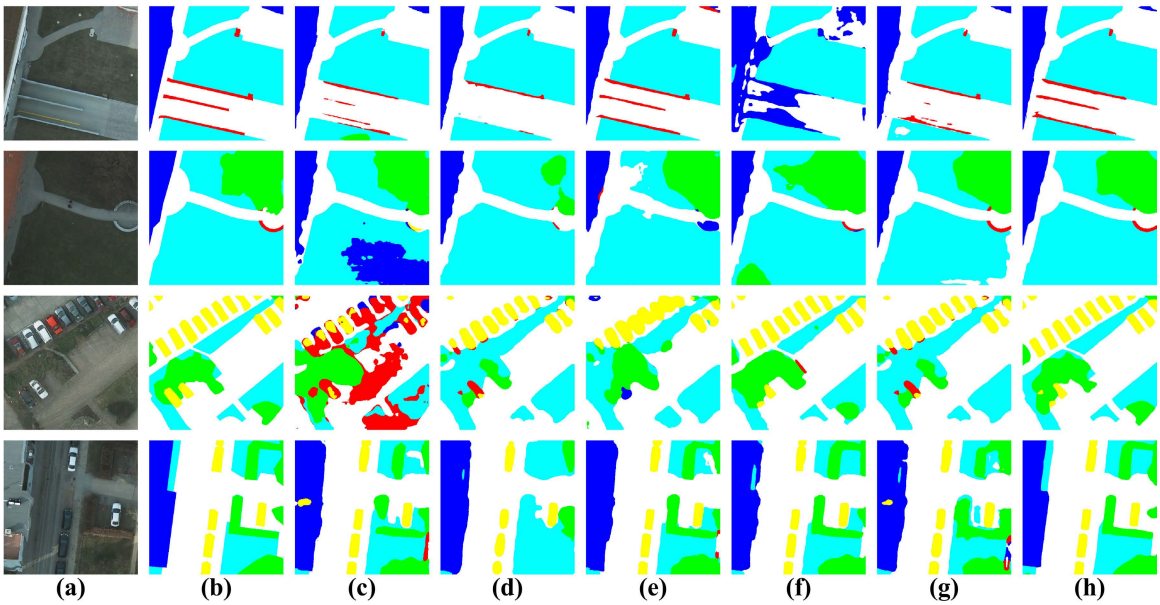


Fig. 13. Qualitative comparison of ablation study results on the ISPRS Potsdam test set. (a) Input image. (b) Ground truth. (c) FRMM. (d) SCFM. (e) FRMM + SCFM. (f) FRMM + MFRM. (g) SCFM + MFRM. (h) FSDPNet.

TABLE V

ABLATION STUDY OF MODEL STRUCTURE VARIANTS ON ISPRS POTSDAM DATASET. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER FIVE RUNS. THE BEST VALUE IN EACH COLUMN IS HIGHLIGHTED IN BOLD

FRMM	SCFM	MFRM	Imp-suf	Building	Low-veg	Tree	Car	Clutter	mIoU (%)
✓	×	×	85.08 ± 0.17	92.55 ± 0.14	74.20 ± 0.19	75.47 ± 0.23	90.39 ± 0.21	38.18 ± 0.42	75.98 ± 0.11
×	✓	×	86.27 ± 0.23	94.46 ± 0.10	73.43 ± 0.16	74.74 ± 0.18	91.02 ± 0.27	39.07 ± 0.33	76.50 ± 0.13
×	×	✓	84.74 ± 0.28	92.79 ± 0.16	75.04 ± 0.14	76.68 ± 0.13	92.17 ± 0.39	37.86 ± 0.24	76.55 ± 0.19
✓	✓	×	87.52 ± 0.21	94.30 ± 0.13	77.78 ± 0.24	79.08 ± 0.12	90.75 ± 0.20	**40.44** ± 0.20	78.31 ± 0.15
✓	×	✓	85.47 ± 0.17	93.20 ± 0.12	78.25 ± 0.17	77.24 ± 0.18	93.01 ± 0.22	38.56 ± 0.32	77.62 ± 0.13
×	×	✓	86.71 ± 0.20	93.43 ± 0.09	76.67 ± 0.13	78.01 ± 0.16	91.64 ± 0.14	39.68 ± 0.26	77.69 ± 0.12
✓	✓	✓	**88.29** ± 0.15	**93.91** ± 0.07	**78.35** ± 0.10	**81.22** ± 0.13	**93.76** ± 0.10	38.68 ± 0.15	**79.04** ± 0.08

shown in Tables V–VII, introducing only the MFRM yields mIoU scores of 76.55%, 72.42%, and 54.66% on the three datasets, with particularly significant improvements for water and large agricultural areas in LoveDA. MFRM effectively mitigates high-frequency attenuation during upsampling and better recovers fine spatial structures across scales. Qual-

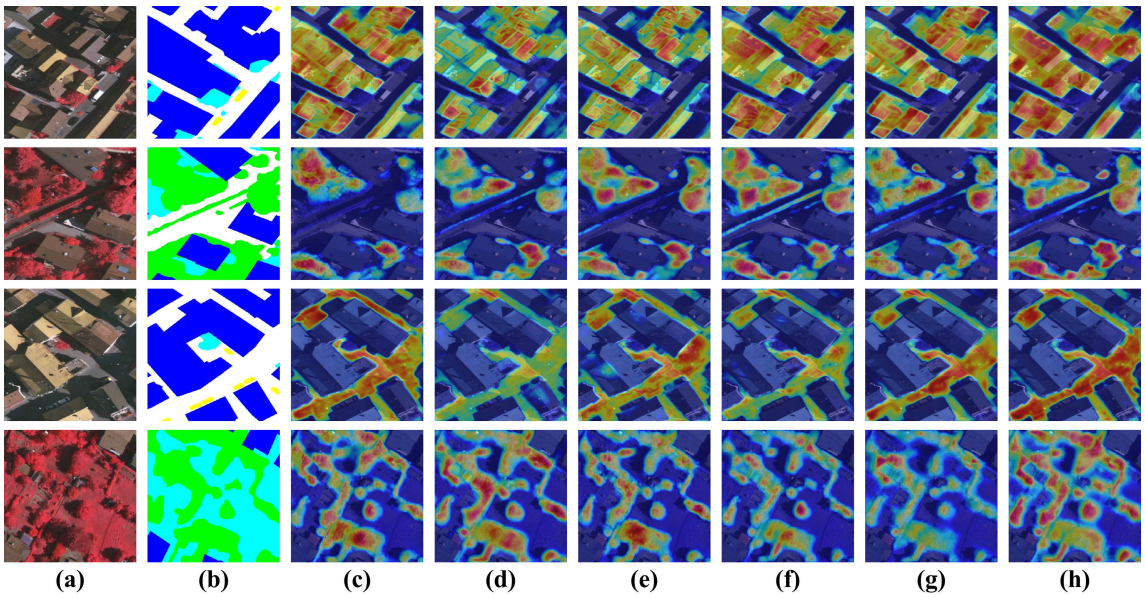


Fig. 14. Feature visualization of ablation study results on the ISPRS Vaihingen test set. (a) Input image. (b) Ground truth. (c) FRMM. (d) SCFM. (e) FRMM + SCFM. (f) FRMM + MFRM. (g) SCFM + MFRM. (h) FSDPNet.

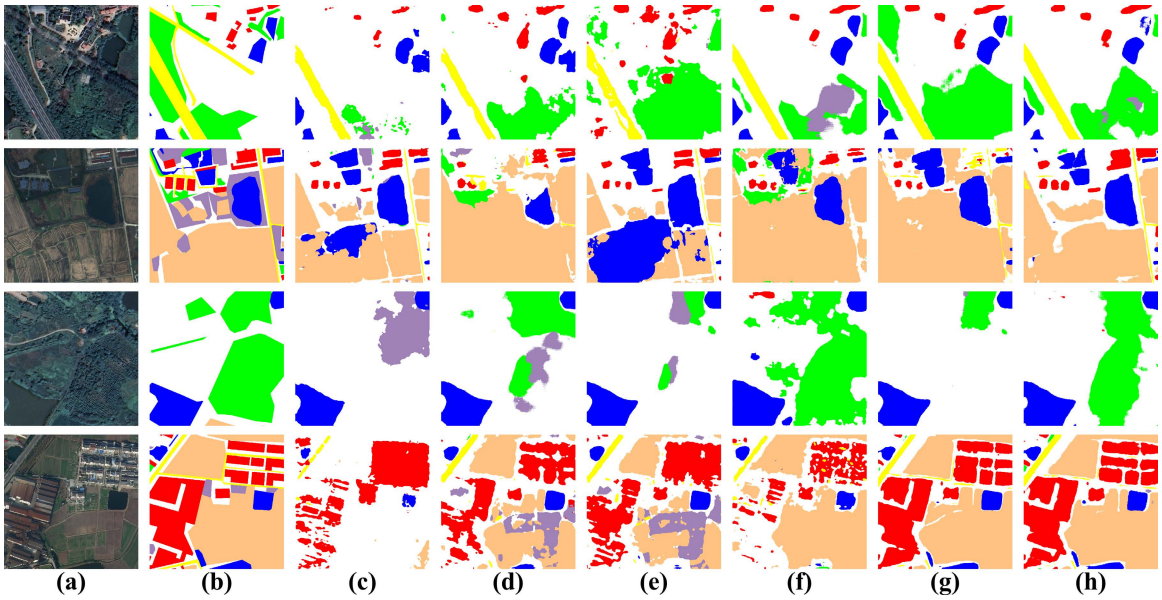


Fig. 15. Qualitative comparison of ablation study results on the LoveDA test set. (a) Input image. (b) Ground truth. (c) FRMM. (d) SCFM. (e) FRMM + SCFM. (f) FRMM + MFRM. (g) SCFM + MFRM. (h) FSDPNet.

itative results (see Figs. 13 and 14) verify that MFRM enables sharp and continuous boundary predictions, especially under complex backgrounds and occlusions. Feature heatmap comparisons (see Fig. 15) further demonstrate that MFRM strengthens feature aggregation in structural regions and enhances foreground-background contrast. When all three modules are integrated, our FSDPNet achieves the highest mIoU on all datasets, 79.04%, 77.76%, and 59.08%, verifying the complementary and synergistic effects of the proposed components.

4) Sensitivity to Loss Weighting: To further investigate the impact of different loss components on model performance, we conduct sensitivity analysis on three loss terms (cross entropy, frequency consistency, and semantic alignment losses), with their weights controlled by λ_1, λ_2, and λ_3, respectively. The results (see Table IX) on ISPRS Vaihingen indicate that model performance is sensitive to the weighting scheme. When only the cross-entropy loss is used ($\lambda_1 = 1$ and $\lambda_2 = \lambda_3 = 0$), the mIoU is 74.79%, with clear limitations in boundary and high-frequency detail recovery. Assigning equal weights ($\lambda_1 = \lambda_2 = \lambda_3 = 0.33$) leads to a slight improvement (mIoU 74.90%), while introducing moderate weighting ($\lambda_1 = 0.5$, $\lambda_2 = 0.3$, and $\lambda_3 = 0.2$) achieves the best performance (mIoU 77.76%). This balanced weighting most effectively improves the segmentation of key classes such as buildings and low-vegetation, confirming the benefit of jointly optimizing multiscale consistency and semantic alignment.

TABLE VI

ABLATION STUDY OF MODEL STRUCTURE VARIANTS ON ISPRS VAIHINGEN DATASET. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER FIVE RUNS. THE BEST VALUE IN EACH COLUMN IS HIGHLIGHTED IN BOLD

FRMM	SCFM	MFRM	Imp-suf	Building	Low-veg	Tree	Car	Clutter	mIoU (%)
✓	×	×	80.25 ± 0.22	90.62 ± 0.18	67.83 ± 0.24	78.36 ± 0.17	75.83 ± 0.32	38.27 ± 0.19	71.86 ± 0.17
×	✓	×	79.12 ± 0.19	88.47 ± 0.22	65.19 ± 0.38	75.14 ± 0.22	77.46 ± 0.16	39.91 ± 0.23	70.88 ± 0.13
×	×	✓	77.46 ± 0.33	89.23 ± 0.29	69.28 ± 0.21	77.82 ± 0.19	79.12 ± 0.29	41.63 ± 0.34	72.42 ± 0.28
✓	✓	×	83.91 ± 0.17	91.78 ± 0.20	70.57 ± 0.17	81.59 ± 0.15	80.35 ± 0.14	38.58 ± 0.11	74.46 ± 0.10
✓	×	✓	86.33 ± 0.12	93.15 ± 0.17	71.42 ± 0.19	79.47 ± 0.17	79.69 ± 0.16	39.72 ± 0.11	74.96 ± 0.11
×	×	✓	85.07 ± 0.09	94.56 ± 0.20	72.91 ± 0.14	82.25 ± 0.08	80.54 ± 0.18	42.15 ± 0.12	76.25 ± 0.08
✓	✓	✓	$\mathbf{88.27} \pm 0.08$	$\mathbf{95.05} \pm 0.07$	$\mathbf{74.28} \pm 0.13$	$\mathbf{83.92} \pm 0.13$	$\mathbf{81.98} \pm 0.10$	$\mathbf{43.04} \pm 0.15$	$\mathbf{77.76} \pm 0.07$

TABLE VII

ABLATION STUDY OF MODEL STRUCTURE VARIANTS ON LOVEDA DATASET. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER FIVE RUNS. THE BEST VALUE IN EACH COLUMN IS HIGHLIGHTED IN BOLD

FRMM	SCFM	MFRM	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU (%)
✓	×	×	52.36 ± 0.23	58.39 ± 0.18	52.67 ± 0.24	75.62 ± 0.34	$\mathbf{38.57} \pm 0.21$	45.73 ± 0.17	62.33 ± 0.15	55.10 ± 0.14
×	✓	×	51.81 ± 0.19	56.21 ± 0.18	51.91 ± 0.19	76.19 ± 0.16	36.92 ± 0.14	43.18 ± 0.22	61.17 ± 0.19	53.91 ± 0.13
×	×	✓	52.92 ± 0.14	57.06 ± 0.16	53.23 ± 0.17	78.45 ± 0.27	30.15 ± 0.27	46.95 ± 0.25	63.89 ± 0.13	54.66 ± 0.09
✓	✓	×	55.47 ± 0.18	61.33 ± 0.13	56.14 ± 0.22	79.03 ± 0.21	38.41 ± 0.18	49.62 ± 0.20	65.42 ± 0.17	57.92 ± 0.11
✓	×	✓	57.63 ± 0.19	60.88 ± 0.16	54.80 ± 0.18	79.28 ± 0.15	35.08 ± 0.19	47.29 ± 0.17	65.21 ± 0.21	57.17 ± 0.08
×	×	✓	56.88 ± 0.20	59.74 ± 0.14	57.36 ± 0.20	78.84 ± 0.15	37.66 ± 0.18	48.07 ± 0.19	63.75 ± 0.18	57.47 ± 0.10
✓	✓	✓	$\mathbf{58.27} \pm 0.14$	$\mathbf{62.77} \pm 0.12$	$\mathbf{58.35} \pm 0.15$	$\mathbf{80.60} \pm 0.13$	37.24 ± 0.17	$\mathbf{50.26} \pm 0.16$	$\mathbf{66.08} \pm 0.09$	$\mathbf{59.08} \pm 0.07$

TABLE VIII

ABLATION STUDY OF FREQUENCY TRANSFORM TYPES IN FRMM ON THE LOVEDA DATASET. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER FIVE RUNS. THE BEST VALUE IN EACH COLUMN IS HIGHLIGHTED IN BOLD

Transform	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU (%)	FPS
Baseline	56.18±0.16	60.50±0.13	55.13±0.17	78.24±0.19	33.07±0.22	47.90±0.20	62.41±0.14	56.49±0.09	**33.0**
DCT (Ours)	**58.27**±0.14	**62.77**±0.12	**58.35**±0.15	**80.60**±0.13	**37.24**±0.17	**50.26**±0.16	**66.08**±0.09	**59.08±0.07**	28.7
FFT	57.46±0.18	61.91±0.13	57.43±0.16	79.82±0.16	36.18±0.15	49.32±0.18	65.22±0.12	58.05±0.11	26.9
Wavelet (Haar)	57.83±0.15	61.52±0.14	56.75±0.13	79.39±0.17	34.16±0.13	48.78±0.14	64.90±0.10	57.62±0.08	25.4
Learnable Filter	58.09±0.17	62.01±0.11	57.99±0.13	80.01±0.15	36.97±0.14	49.86±0.12	65.87±0.09	58.53±0.09	21.5

TABLE IX

ABLATION STUDY OF LOSS WEIGHTING COEFFICIENTS ON ISPRS VAIHINGEN DATASET. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER FIVE RUNS. THE BEST VALUE IN EACH COLUMN IS HIGHLIGHTED IN BOLD

λ_1	λ_2	λ_3	Imp-suf	Building	Low-veg	Tree	Car	Clutter	mIoU (%)
1	0	0	84.78 ± 0.13	92.42 ± 0.12	71.44 ± 0.16	81.18 ± 0.11	75.53 ± 0.20	43.36 ± 0.13	74.79 ± 0.11
0.33	0.33	0.33	84.42 ± 0.17	91.99 ± 0.21	71.12 ± 0.13	80.85 ± 0.21	76.19 ± 0.15	44.81 ± 0.13	74.90 ± 0.10
0.8	0.1	0.1	85.39 ± 0.19	94.63 ± 0.09	72.51 ± 0.09	82.47 ± 0.13	78.08 ± 0.11	$\mathbf{45.94} \pm 0.08$	76.50 ± 0.07
0.6	0.2	0.2	86.40 ± 0.13	93.66 ± 0.11	70.75 ± 0.15	81.10 ± 0.10	71.38 ± 0.14	43.22 ± 0.14	74.42 ± 0.08
0.4	0.3	0.3	86.69 ± 0.08	94.07 ± 0.13	72.98 ± 0.18	83.28 ± 0.13	80.81 ± 0.10	42.73 ± 0.11	76.76 ± 0.10
0.5	0.3	0.2	$\mathbf{88.27} \pm 0.06$	$\mathbf{95.05} \pm 0.07$	$\mathbf{74.28} \pm 0.10$	$\mathbf{83.92} \pm 0.07$	$\mathbf{81.98} \pm 0.08$	43.04 ± 0.09	$\mathbf{77.76} \pm 0.03$

5) Impact of Frequency Transform Choice: To further validate the selection of the DCT in the FRMM module, we performed a transform-type ablation on the LoveDA dataset. As reported in Table VIII, we compared the baseline (no frequency modeling), DCT, fast Fourier transform (FFT), wavelet transform (Haar), and a learnable frequency filter. The results show that all frequency-domain methods yield clear performance gains over the baseline, confirming the importance of spectral decomposition. DCT achieves the highest mIoU (59.08%) and maintains favorable inference speed (28.7 frames/s), outperforming FFT (58.05% mIoU and 26.9 frames/s), wavelet (57.62% mIoU and 25.4 frames/s), and learnable filters (58.53% mIoU and 21.5 frames/s). DCT offers an optimal tradeoff between accuracy and efficiency, attributed to its compact energy representation and purely real-valued operations. These results substantiate our design choice, demonstrating that DCT is effective and efficient for frequency-domain modeling in RSI segmentation.

V. DISCUSSION

In this section, we further analyze the effectiveness and limitations of the proposed FSDPNet through additional ablation studies and failure case analysis. Cross-domain generalization results are summarized in Table XIII.

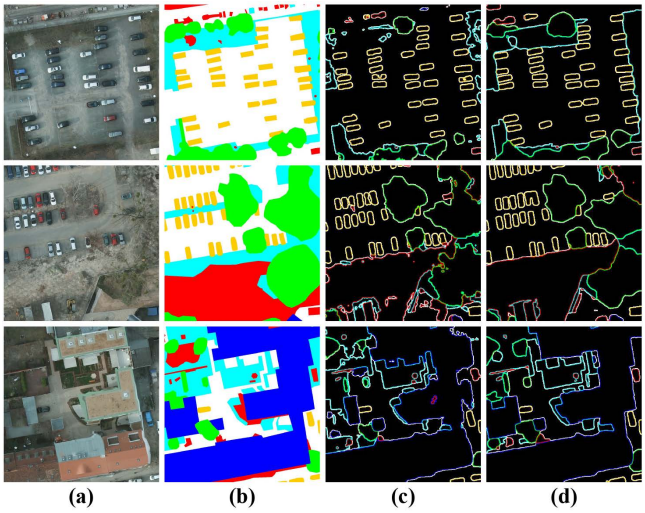


Fig. 16. Qualitative comparison of segmentation boundaries with and without SAM priors in diverse remote sensing scenes. (a) Input image. (b) Ground truth. (c) Model without SAM prior. (d) Model with SAM prior.

TABLE X
QUANTITATIVE EVALUATION OF SAM PRIOR ROBUSTNESS ON THE POTSDAM DATASET

Scene Type	Method	OA (%)	mIoU (%)	mF1 (%)
Simple	w/o SAM Prior	91.20	82.53	85.13
	w SAM Prior	93.41	86.37	88.92
Complex	w/o SAM Prior	88.58	77.08	80.64
	w SAM Prior	87.74	75.32	78.38

A. Analysis of SAM Prior

To further quantify the impact of SAM-derived priors on segmentation performance, we conduct both qualitative and quantitative analyses across diverse remote sensing scenarios. As illustrated in Fig. 16, in simple scenes characterized by clear object boundaries, regular shapes, and minimal background interference, such as dense vehicle arrangements in parking lots, the incorporation of SAM priors markedly improves edge continuity, reduces boundary ambiguity, and enhances the preservation of fine details.

Corresponding quantitative results on the Potsdam dataset, as shown in Table X, demonstrate that, for such scenes, introducing the SAM prior yields significant improvements: OA rises from 91.20% to 93.41%, mIoU from 82.53% to 86.37%, and m$F1$ from 85.13% to 88.92%. In contrast, for complex urban environments with substantial background clutter, ambiguous textures, or overlapping and occluded structures (e.g., mixed rooftops and cast shadows), the reliability of SAM priors decreases, occasionally leading to spurious or missing boundaries.

In these cases, as shown in Table X, applying the SAM prior may slightly reduce accuracy: OA drops from 88.58% to 87.74%, mIoU from 77.08% to 75.32%, and m$F1$ from 80.64% to 78.38%. These indicate that while SAM priors are beneficial in structured, low-interference settings, their effectiveness diminishes in highly complex or ambiguous regions, where unreliable priors can degrade the outcome.

B. Analysis of Loss Function

To further evaluate the contribution of our proposed loss function, we apply the unified loss configuration to several state-of-the-art frequency-based methods, including SFFNet, FDNet, AFENet, FGSwIN, and FreDNet, and compare their performance with our FSDPNet on the ISPRS Vaihingen dataset. As presented in Table XI, consistent application of our loss function substantially narrows the gap in experimental settings, thereby highlighting the structural differences among the tested methods. The results demonstrate that our loss design enables all compared models to achieve stable improvements across categories; however, FSDPNet consistently outperforms the other methods by a substantial margin in both per-class IoU and overall scores. For example, FSDPNet achieves an mIoU of 77.76% and m$F1$ of 85.94%, significantly surpassing the best baseline (FGSwIN, mIoU 69.93%, m$F1$% and 78.15%). The improvement is particularly clear in challenging categories such as low-vegetation, tree, car, and clutter, where FSDPNet yields noticeable gains (e.g., up to 14.58% higher IoU for low-veg and 11.01% for car compared with the strongest competing method). These results clearly indicate that while careful tuning of the loss function is important, our observed performance gains primarily originate from the architectural contributions of FSDPNet, rather than from loss parameter sensitivity alone.

C. Analysis of Failure Cases

To further clarify the limitations of FSDPNet, we analyzed representative failure cases, as illustrated in Fig. 17. Most errors are concentrated at category boundaries, around small or fragmented objects, and in regions with strong spectral or spatial similarity between classes. For example, the model may partially miss vehicles and narrow roads in ISPRS Potsdam, confuse roads with parking lots or railways in ISPRS Vaihingen, and struggle with sparsely distributed buildings and roads surrounded by vegetation in LoveDA. To provide a quantitative perspective, we assessed all validation images for three typical failure types: boundary confusion, small object omission, and interclass misclassification.

As reported in Table XII, each type affects fewer than 9% of images, with mean pixel-level error ratios below 4.5%. These results confirm the overall robustness of FSDPNet, while highlighting that most residual errors occur in regions that are inherently ambiguous or involve fine-scale details. The main causes include limited spatial resolution for small targets, strong interclass similarities, and insufficient structural priors. Future work could further alleviate these issues by introducing high-resolution auxiliary data (such as DSM or LiDAR), structure-aware modules, or domain adaptation techniques to improve feature discrimination and generalization in complex scenes.

D. Object-Size-Based Segmentation Performance

The experimental results on the ISPRS Vaihingen dataset demonstrate that the proposed FSDPNet consistently outperforms the baseline (UNetFormer), particularly in the segmentation of small objects and fine-grained structures.

TABLE XI

QUANTITATIVE COMPARISON OF PER-CLASS IOU AND OVERALL METRICS FOR DIFFERENT FREQUENCY DOMAIN-BASED SEGMENTATION METHODS ON THE ISPRS VAIHINGEN DATASET, ALL TRAINED WITH THE PROPOSED LOSS FUNCTION

Methods	Imp-suf	Building	Low-veg	Tree	Car	Clutter	OA (%)	mIoU (%)	mF1 (%)
SFFNet [80]	80.41	86.87	59.70	67.03	61.32	29.19	84.56	64.75	74.19
FDNet [81]	82.28	88.95	63.48	71.22	66.41	31.65	86.72	67.67	76.92
AFENet [82]	78.16	87.11	58.51	66.29	64.20	28.48	83.89	63.79	72.64
FGSwin [83]	84.36	90.28	65.07	73.15	68.54	34.16	87.93	69.93	78.15
FreDNet [84]	81.79	88.24	62.09	70.33	65.09	31.98	86.41	66.62	76.07
FSDPNet (Ours)	**88.27**	**95.05**	**74.28**	**83.92**	**81.98**	**43.04**	**92.57**	**77.76**	**85.94**

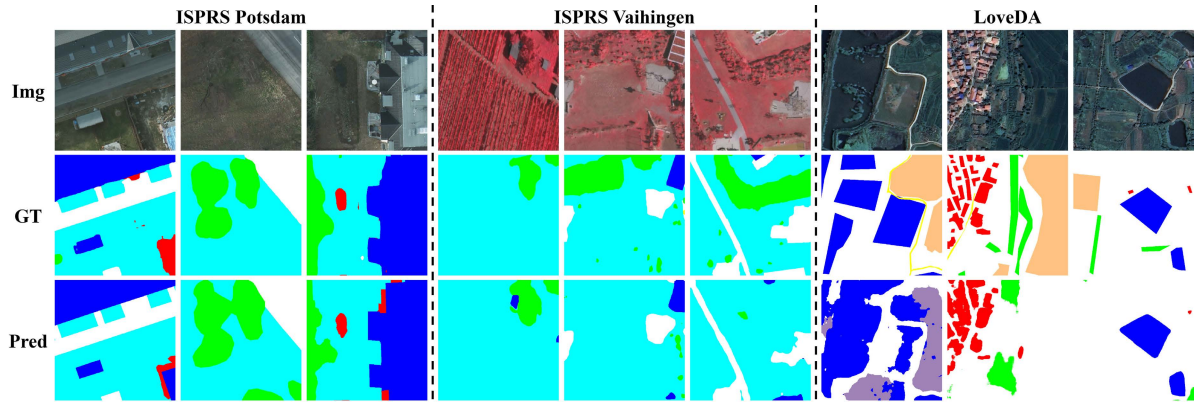


Fig. 17. Representative failure cases of FSDPNet on the ISPRS Potsdam, ISPRS Vaihingen, and LoveDA datasets.

TABLE XII

QUANTITATIVE ANALYSIS OF FAILURE CASE FREQUENCIES (% OF VALIDATION IMAGES AND MEAN PIXEL-LEVEL ERROR RATIO) ON THREE DATASETS

Dataset	Failure Type	Affected Images	Pixel Error Ratio
Potsdam	Boundary confusion	8.7	4.2
	Small object omission	6.0	2.3
	Inter-class misclassification	5.3	2.8
Vaihingen	Boundary confusion	7.9	3.9
	Small object omission	5.5	2.1
	Inter-class misclassification	4.8	2.4
LoveDA	Boundary confusion	8.1	4.4
	Small object omission	7.3	3.0
	Inter-class misclassification	8.8	3.5

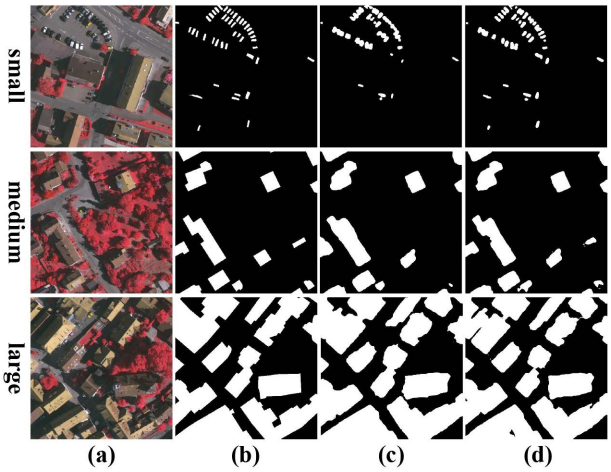


Fig. 18. Visual comparison of segmentation results for different object size groups on the Vaihingen dataset. (a) Input image. (b) Ground truth. (c) Baseline (UNetFormer). (d) FSDPNet (Ours).

As summarized in Table XIV, FSDPNet achieves an IoU of 61.87% ± 0.69% for small objects, exceeding the baseline by 7.14 percentage points (54.73% ± 0.81%). Notable improvements are also observed for medium-sized objects (77.42% ± 0.51% versus 75.21% ± 0.63%), while performance on large objects remains comparable (84.97% ± 0.21% versus 84.34% ± 0.18%). These results substantiate the effectiveness of our method in recognizing small-scale and intricate structures, which are typically challenging for conventional approaches. The qualitative comparisons in Fig. 18 further support these findings. For both small and medium object groups, FSDPNet recovers more complete masks and yields smoother contours than the baseline. In particular, our method effectively distinguishes small building footprints that are easily missed or fragmented by the baseline. For large objects, both methods achieve high accuracy, indicating the principal advantage lies with challenging, small-scale objects. This object-size-based evaluation explicitly confirms that FSDPNet mitigates the issue of small-object omission, reinforcing the central claim regarding its capability for fine-grained segmentation in high-resolution remote sensing imagery.

E. Cross-Domain Generalization Under Domain Shift

To comprehensively evaluate the robustness of FSDPNet in the presence of domain shift, we conduct a series of cross-domain experiments using two widely adopted

TABLE XIII
CROSS-DOMAIN SEGMENTATION PERFORMANCE (%) UNDER DOMAIN SHIFT. ALL MODELS ARE TRAINED ON THE SOURCE DOMAIN AND EVALUATED ON THE TARGET DOMAIN

Methods	Potsdam→Vaihingen		Vaihingen→Potsdam		LoveDA Urban→Rural	
	mIoU	mF1	mIoU	mF1	mIoU	mF1
FarSeg [65]	58.54	67.43	55.21	66.97	42.08	54.73
Segmenter [68]	60.33	68.29	58.51	69.81	45.92	57.80
RS3Mamba [73]	62.12	70.41	60.00	70.22	46.87	58.29
UNetMamba [75]	63.45	71.76	61.84	71.65	48.03	59.84
FSDPNet (Ours)	**65.97**	**73.08**	**64.15**	**73.52**	**52.16**	**63.00**

TABLE XIV
OBJECT-SIZE-BASED SEGMENTATION IOU (%) ON THE VAIHINGEN DATASET. RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER FIVE RUNS

Methods	Small	Medium	Large
Baseline	54.73 ± 0.81	75.21 ± 0.63	84.34 ± 0.18
FSDPNet (Ours)	**61.87** ± 0.69	**77.42** ± 0.51	**84.97** ± 0.21

transfer scenarios: Potsdam→Vaihingen, Vaihingen→Potsdam, and LoveDA Urban→Rural. In each case, models are trained exclusively on the source domain and directly tested on the target domain, without any adaptation. As shown in Table XIII, FSDPNet consistently outperforms representative CNN-, Transformer-, and Mamba-based baselines in all transfer settings. FSDPNet achieves clear performance margins, improving mIoU by 2%–4% over the best competitors, thereby verifying its superior generalization under both intracity (Potsdam↔Vaihingen) and urban–rural (LoveDA) scenarios. We attribute this robustness to the frequency–spatial dual-path architecture, which enhances the model's ability to retain structural cues and suppresses semantic drift in novel domains. These results provide compelling evidence for the practical generalization capability of FSDPNet under real-world domain shifts.

VI. CONCLUSION

This article introduces FSDPNet, a novel FSDPNet framework for RSI semantic segmentation. By coupling frequency-domain encoding, multiscale frequency-driven reconstruction, and semantic-guided fusion, FSDPNet effectively preserves fine structural details and enhances global semantic consistency. The proposed network leverages frequency-aware modules to mitigate high-frequency information loss, incorporates SAM-derived priors for semantic alignment, and utilizes multiscale feature aggregation to improve both boundary delineation and object integrity. Comprehensive experiments on several benchmarks demonstrate significant improvements over state-of-the-art methods, especially in challenging scenarios with complex structures and ambiguous boundaries. Ablation studies further confirm the complementary strengths of each module and the overall robustness of the framework. In future work, we will explore structure-aware learning and multimodal data integration to further advance robust segmentation in diverse real-time remote sensing applications.

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